Past Problems

for (MATH203) Applied Linear Algebra

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POSTECH

Problem 1. (2018 Fall POSTECH Applied Linear Algebra). In this problem, V is a vector space. Determine whether each of the following statements is true or false.

(1) There exists a linearly independent subset $S \subseteq V$ that contains the zero vector $\mathbf{0}$.
(2) If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is invertible, then the columns of $\mathbf{A}$ form a basis for $\mathbb{R}^n$.
(3) Every projection matrix is invertible.
(4) $\mathrm{Null}(\mathbf{A})^{\perp} = \mathrm{Col}(\mathbf{A})$.
(5) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V, and let $T\colon V \to W$ be a linear transformation. If

$$T(\mathbf{v}_i) = \mathbf{w}_i, \qquad 1 \le i \le n,$$

for given vectors $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$, then T is uniquely determined.

Problem 2. (2018 Fall POSTECH Applied Linear Algebra). For each statement, prove it or disprove it by providing a counterexample.

(1) Consider a linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \qquad \mathbf{A} \in \mathbb{R}^{m\times n}, \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{b} \in \mathbb{R}^m.$$

If there are more variables than equations, $m < n$, then the system always has a solution.
(2) If $\mathbf{A} \in \mathbb{R}^{n\times n}$ satisfies

$$\mathbf{A}^2 = \mathbf{A} \qquad \text{and} \qquad \mathrm{rk}(\mathbf{A}) = n,$$

then $\mathbf{A} = \mathbf{I}_n$.
(3) The null space of $\mathbf{A}$ is equal to the null space of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$.

Problem 3. (2018 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

(a) Find $\mathbf{A}^{-1}$.
(b) Compute $(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}$.

Problem 4. (2018 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

(a) Find a right inverse $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{I}_2.$$

(b) Using the matrix $\mathbf{B}$ obtained in part (a), compute $(\mathbf{BA})^{100}$.

Problem 5. (2018 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 4 & 4 \\ 4 & 4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Find the orthogonal projection $\mathbf{b}_c$ of $\mathbf{b}$ onto $\mathrm{Col}(\mathbf{A})$.

Problem 6. (2018 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{pmatrix}.$$

(a) Find an LU decomposition $\mathbf{A} = \mathbf{LU}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.
(b) Find a basis for $\mathrm{Col}(\mathbf{B})$.
(c) Find $\mathrm{Null}(\mathbf{B})$.
(d) Write the complete solution in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ for

$$\mathbf{C}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix},$$

where $\mathbf{x}_p$ is a particular solution and $\mathbf{x}_n \in \mathrm{Null}(\mathbf{C})$.

Problem 7. (2018 Fall POSTECH Applied Linear Algebra). Let $V = M_2(\mathbb{R})$, the vector space of 2×2 real matrices with the usual addition and scalar multiplication.

(a) Show that

$$\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

form a basis for V.

(b) Find the matrix representation of the linear transformation

$$T\colon V \to V, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a+d & c \\ b & a-d \end{pmatrix},$$

with respect to the ordered basis $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$.

Problem 8. (2018 Fall POSTECH Applied Linear Algebra). Let

$$P_k = \left\{ a_k t^k + a_{k-1} t^{k-1} + \cdots + a_1 t + a_0 \mid a_0, \ldots, a_k \in \mathbb{R} \right\},$$

with the usual polynomial addition and scalar multiplication. Use the ordered basis

$$(1, t, t^2, \ldots, t^k)$$

for P_k.

(a) Find the matrix representation $\mathbf{A}$ of the differentiation map

$$D \colon P_2 \to P_1, \qquad D(f) = \frac{\mathrm{d}f}{\mathrm{d}t}.$$

(b) Find the matrix representation $\mathbf{B}$ of the integration map

$$T \colon P_1 \to P_2, \qquad (Tf)(t) = \int_0^t f(s)\, \mathrm{d}s.$$

Problem 9. (2018 Fall POSTECH Applied Linear Algebra). Determine whether each of the following statements is true or false.

(1) $\mathbf{A}$ and $\mathbf{A}^\mathsf{T}\mathbf{A}$ have the same rank for every rectangular matrix $\mathbf{A}$.

(2) The only possible eigenvalues of a symmetric orthogonal matrix are 1 and -1.

(3) For an $n \times n$ matrix $\mathbf{A}$ and a scalar $k \in \mathbb{R}$,

$$\det(k\mathbf{A}) = k \det(\mathbf{A}).$$

(4) For an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

(5) If there is a basis consisting of eigenvectors of a matrix, then the matrix is diagonalizable.

(6) Every diagonalizable matrix has an orthonormal basis consisting of eigenvectors.

Problem 10. (2018 Fall POSTECH Applied Linear Algebra). Compute the determinants of the following matrices.

(a) Use cofactor expansion to find the determinant of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -1 \\ -1 & 1 & 3 \end{pmatrix}.$$

(b) Transform the matrix into upper triangular form using elementary row operations, and compute the determinant of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 2 & 4 & 0 \end{pmatrix}.$$

Problem 11. (2018 Fall POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be real square matrices. Prove that if $\mathbf{A}$ and $\mathbf{B}$ are orthogonally similar, then they have the same singular values.

Problem 12. (2018 Fall POSTECH Applied Linear Algebra). Write down as many pairwise nonsimilar 4×4 matrices as possible whose only eigenvalue is 2.

Problem 13. (2018 Fall POSTECH Applied Linear Algebra). Consider $\mathbb{R}^4$ with the standard inner product and let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \subseteq \mathbb{R}^4,$$

where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 3 \\ 2 \\ -3 \end{pmatrix}, \qquad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$$

(a) Find a subset of S that is a basis for $\mathrm{Span}(S)$ and includes $\{\mathbf{v}_1, \mathbf{v}_2\}$.

(b) Apply the Gram–Schmidt process to the subset found in part (a) and obtain an orthonormal basis for $\mathrm{Span}(S)$.

Problem 14. (2018 Fall POSTECH Applied Linear Algebra). Identify and sketch the curve

$$x^2 + xy + y^2 = 1.$$

Justify your identification with an explicit equation. Indicate all x-intercepts and y-intercepts, and label all coordinate axes.

Problem 15. (2018 Fall POSTECH Applied Linear Algebra). Solve the initial value problem

$$\mathbf{u}'(t) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \mathbf{u}(t), \qquad \mathbf{u}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Problem 16. (2018 Fall POSTECH Applied Linear Algebra). Use a similarity transformation to find the matrix $\mathbf{A} \in \mathbb{R}^{3\times 3}$ representing reflection across the plane

$$x + y + z = 0.$$

Problem 17. (2018 Fall POSTECH Applied Linear Algebra). Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 3 \\ 0 & 0 & 5 \end{pmatrix}.$$

Problem 18. (2019 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}.$$

(a) Find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{P}$ is diagonal.

(b) Find the characteristic polynomial of $\mathbf{A}$.

(c) Show that $\mathbf{A}^{-1}$ can be expressed as

$$a\mathbf{A}^2 + b\mathbf{A} + c\mathbf{I}_3$$

for some $a, b, c \in \mathbb{R}$, and find a, b, c.

(d) Find $e^{\mathbf{A}}$.

Problem 19. (2019 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Find all eigenvalues of $\mathbf{A}$.
(b) Find all eigenvectors of $\mathbf{A}$.
(c) Determine whether $\mathbf{A}$ is diagonalizable. Explain why.
(d) Find the Jordan form of $\mathbf{A}$.
(e) Find a Jordan basis of $\mathbf{A}$.
(f) Find $\det(e^{\mathbf{A}})$.
(g) Calculate $e^{\mathbf{A}}$.
(h) Solve

$$\mathbf{y}' = \mathbf{A}\mathbf{y}, \qquad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}.$$

Problem 20. (2019 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{D} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix},$$

where $\mathbf{A}$ and $\mathbf{C}$ are square matrices and $\mathbf{0}$ is a zero matrix of compatible size. Prove that

$$\det(\mathbf{D}) = \det(\mathbf{A}) \det(\mathbf{C}).$$

Problem 21. (2019 Spring POSTECH Applied Linear Algebra). Let (g_k) be a sequence satisfying

$$g_{k+2} = g_{k+1} - g_k, \qquad g_1 = 1, \qquad g_2 = 2.$$

(a) Find g_{300}.
(b) Determine whether the sequence (g_k) converges. In either case, explain why.

Problem 22. (2019 Spring POSTECH Applied Linear Algebra). Calculate

$$\det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Problem 23. (2019 Spring POSTECH Applied Linear Algebra). Let

$$f(x) = \det \begin{pmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & x & x^2 & x^3 \end{pmatrix}.$$

Find all zeros of $f(x)$ and verify your answer.

Problem 24. (2019 Spring POSTECH Applied Linear Algebra). Let $\mathbf{M} \in \mathbb{R}^{n\times n}$ be invertible, and let $\mathbf{v}_1, \ldots, \mathbf{v}_l \in \mathbb{R}^n$ be linearly independent. Prove that

$$\mathbf{M}\mathbf{v}_1, \ldots, \mathbf{M}\mathbf{v}_l$$

are also linearly independent.

Problem 25. (2019 Spring POSTECH Applied Linear Algebra). Prove that the two matrices

$$\begin{pmatrix} \mathbf{A}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mathbf{A}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_1 \end{pmatrix}$$

are similar, where $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are square matrices of sizes m, n, l, respectively.

Problem 26. (2019 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1+i \\ 0 & 1 & 1-i \\ 0 & 0 & i \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$.
(b) Find the eigenvectors of $\mathbf{A}$.
(c) Find a matrix $\mathbf{S}$ such that $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ is diagonal.

Problem 27. (2019 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ real symmetric matrix, and let λ be an eigenvalue of $\mathbf{A}$ with algebraic multiplicity m. Show that

$$\dim \operatorname{Null}(\mathbf{A} - \lambda \mathbf{I}_n) = m.$$

Problem 28. (2019 Spring POSTECH Applied Linear Algebra).

(1) State and prove Schur's Lemma.
(2) State and prove the complex spectral theorem.

Problem 29. (2019 Fall POSTECH Applied Linear Algebra). Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} 1, & i - j = 1 \text{ and } i < n, \\ 0, & \text{otherwise.} \end{cases}$$

Find the sum of all entries of

$$\sum_{k=1}^{2n} \mathbf{A}^k.$$

Problem 30. (2019 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} a & 0 & b \\ a & a & 4 \\ 0 & a & 2 \end{pmatrix}.$$

For what values of a and b does the system

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 2 \\ 4 \\ b \end{pmatrix}$$

have

(1) a unique solution,
(2) a one-parameter family of solutions,
(3) a two-parameter family of solutions,
(4) no solution?

Problem 31. (2019 Fall POSTECH Applied Linear Algebra). Find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & -3 & 0 \\ 1 & -2 & 0 & 4 \\ -1 & 2 & 0 & 4 \\ 1 & 2 & 3 & 0 \end{pmatrix}.$$

Problem 32. (2019 Fall POSTECH Applied Linear Algebra). Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be two bases of a vector space V. If

$$\mathbf{e}_i = \sum_{j=1}^{n} a_{ij}\mathbf{f}_j, \qquad i = 1, \dots, n,$$

show that the matrix $\mathbf{A} = (a_{ij})$ is invertible.

Problem 33. (2019 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 5 & 1 & 1 \\ 1 & 1 & 3 & 0 & -1 \\ 2 & 2 & 12 & 1 & 0 \end{pmatrix}.$$

(a) Find a basis for $\mathrm{Row}(\mathbf{A})$ and a basis for $\mathrm{Col}(\mathbf{A})$.
(b) Consider the restriction

$$\mathbf{A}|_{\mathrm{Row}(\mathbf{A})} \colon \mathrm{Row}(\mathbf{A}) \to \mathrm{Col}(\mathbf{A}).$$

Find its matrix representation with respect to the bases obtained in part (a), and verify that this matrix is invertible.

Problem 34. (2019 Fall POSTECH Applied Linear Algebra). Let $V \subseteq \mathbb{R}^3$ be a two-dimensional subspace.

(a) Show that the reflection of any vector $\mathbf{x} \in \mathbb{R}^3$ across V is

$$(2\mathbf{P} - \mathbf{I}_3)\mathbf{x},$$

where $\mathbf{P}$ is the orthogonal projection onto V. *Hint.* Consider an orthonormal basis of V and extend it to an orthonormal basis of $\mathbb{R}^3$.
(b) Let

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

be the standard basis of $\mathbb{R}^3$. Let V be spanned by

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Find the matrix representation of reflection across V with respect to the standard basis.

Problem 35. (2019 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & -1 \\ 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 & 1 \end{pmatrix}.$$

(a) Find the matrices $\mathbf{P}$, $\mathbf{U}$, and $\mathbf{L}$ in a decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

(b) Solve $\mathbf{Ax} = \mathbf{b}$ for $\mathbf{x}$, where

$$\mathbf{b} = \begin{pmatrix} -2 \\ 3 \\ 5 \\ 4 \\ 1 \end{pmatrix}.$$

Problem 36. (2019 Fall POSTECH Applied Linear Algebra). Find a nonzero matrix $\mathbf{A}$ whose null space is spanned by

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

and whose left null space is spanned by

$$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Problem 37. (2019 Fall POSTECH Applied Linear Algebra). Fit a quadratic function

$$f(t) = c_0 + c_1 t + c_2 t^2$$

to the four data points

$$(-1, 8), \qquad (0, 8), \qquad (1, 4), \qquad (2, 16).$$

Explain why your answer is the best fit.

Problem 38. (2019 Fall POSTECH Applied Linear Algebra). Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{QR},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 39. (2019 Fall POSTECH Applied Linear Algebra). Let P_4 be the vector space of real polynomials of degree at most 4 in the variable t. The map

$$T\colon P_4 \to P_4, \qquad T(p(t)) = p(t+1),$$

is linear.

(a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$(1, t+1, t^2+1, t^3+1, t^4+1).$$

(b) Find $\mathbf{A}^{-1}$.

Problem 40. (2019 Fall POSTECH Applied Linear Algebra). Fit a quadratic function

$$f(t) = c_0 + c_1 t + c_2 t^2$$

to the four data points

$$(-1, 8), \qquad (0, 8), \qquad (1, 4), \qquad (2, 16).$$

Explain why your answer is the best fit.

Problem 41. (2019 Fall POSTECH Applied Linear Algebra). Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{QR},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 42. (2019 Fall POSTECH Applied Linear Algebra). For a positive integer n, let S_n be the set of linear combinations of

$$\sin x, \ldots, \sin nx, \cos x, \ldots, \cos nx,$$

that is,

$$S_n = \left\{ \sum_{k=1}^{n} (a_k \sin kx + b_k \cos kx) \;\middle|\; a_k, b_k \in \mathbb{R} \right\}.$$

Then the map

$$D\colon S_n \to S_n, \qquad D(f) = \frac{\mathrm{d}f}{\mathrm{d}x},$$

is linear.

(a) Find the matrix representation of D with respect to the ordered basis

$$(\sin x, \cos x, \sin 2x, \cos 2x, \ldots, \sin nx, \cos nx).$$

(b) Find the determinant and trace of D.

Problem 43. (2019 Fall POSTECH Applied Linear Algebra). Let T be a linear operator on $\mathbb{R}^5$ represented by

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 3 \\ 1 & 2 & -4 & 0 & 3 \\ 1 & 0 & -1 & 1 & 1 \\ 1 & 2 & -3 & 0 & 2 \end{pmatrix}.$$

(a) Show that the characteristic polynomial has the form

$$p_{\mathbf{A}}(\lambda) = -(\lambda - 1)^3(\lambda - a)^2,$$

and find a.

(b) Find bases for the eigenspaces $E_1(\mathbf{A})$ and $E_a(\mathbf{A})$. Here, $E_1(\mathbf{A})$ (respectively, $E_a(\mathbf{A})$) denotes the subspace of $\mathbb{R}^5$ consisting of all eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 1 (respectively, a), together with the zero vector.

(c) Find the Jordan form of $\mathbf{A}$.

Problem 44. (2019 Fall POSTECH Applied Linear Algebra). Factorize

$$\mathbf{A} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1},$$

where $\mathbf{U}$ is unitary and $\mathbf{\Lambda}$ is diagonal.

Problem 45. (2019 Fall POSTECH Applied Linear Algebra). Suppose three cities A, B, and C are centers for trucks. Every month, half of the trucks in A and half of those in B go to C, while the other halves stay where they are. Every month, half of the trucks in C go to A and the other half go to B.

(a) Find the Markov transition matrix acting on the initial-state vector

$$\begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix},$$

where a_0, b_0, c_0 denote the initial numbers of trucks in cities A, B, C, respectively.

(b) If there are always 450 trucks, find the long-run distribution of trucks.

Problem 46. (2019 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}.$$

For what range of a is $\mathbf{A}$ positive definite?

Problem 47. (2019 Fall POSTECH Applied Linear Algebra). Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

Problem 48. (2019 Fall POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix.

(a) Show that $\mathbf{A} = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda \mathbf{I}_n) = (-\lambda)^n$$

and $\mathbf{A}$ is Hermitian.

(b) Show that $\mathbf{A}^n = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda \mathbf{I}_n) = (-\lambda)^n.$$

Hint. Consider the Jordan form.

Problem 49. (2019 Fall POSTECH Applied Linear Algebra). Let λ_0 be an eigenvalue of a square matrix $\mathbf{A}$, and let m be the algebraic multiplicity of λ_0 in the characteristic polynomial of $\mathbf{A}$. Show that

$$\dim E_{\lambda_0}(\mathbf{A}) \leq m.$$

Problem 50. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: the product of triangular matrices is a triangular matrix.

Problem 51. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{D}$ be an $n \times n$ diagonal matrix. Determine whether the following statement is true or false:

$$\mathbf{AD} = \mathbf{DA}$$

for every $n \times n$ matrix $\mathbf{A}$.

Problem 52. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every nonsingular symmetric square matrix $\mathbf{A}$ admits an LDL^{T} factorization, where $\mathbf{L}$ is lower triangular and $\mathbf{D}$ is diagonal.

Problem 53. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a square matrix. Determine whether the following statement is true or false:

$$\dim \mathrm{Col}(\mathbf{A}^2) = \dim \mathrm{Col}(\mathbf{A}).$$

Problem 54. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a rectangular matrix. Determine whether each of the following statements is true or false.

(1) $\mathrm{Null}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \mathrm{Null}(\mathbf{A})$.

(2) $\mathrm{Null}(\mathbf{A}\mathbf{A}^{\mathsf{T}}) = \mathrm{Null}(\mathbf{A})$.

(3) If $\mathbf{A}$ is square, then $\mathrm{Null}(\mathbf{A}^2) = \mathrm{Null}(\mathbf{A})$.

Problem 55. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: if distinct nonzero vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ span a subspace V of $\mathbb{R}^5$, then there exist mutually orthonormal vectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4$ that span V.

Problem 56. (2020 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

$$\mathbf{v}_4 = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_5 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_6 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

Select all true statements.

(1) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ span $\mathbb{R}^4$.
(2) $\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ span $\mathbb{R}^4$.
(3) $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ are linearly independent over $\mathbb{R}$.
(4) $\mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5$ are linearly independent over $\mathbb{R}$.
(5) $\mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5$ span $\mathbb{R}^3$.
(6) None of the above.

Problem 57. (2020 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 & -1 \\ 0 & \frac{8}{3} & \frac{4}{3} \\ 0 & 0 & 0 \end{pmatrix}.$$

Select all true statements.

(1) The vectors

$$\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 8 \\ 4 \end{pmatrix}$$

span $\mathrm{Row}(\mathbf{A})$.
(2) The vectors

$$\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ \frac{4}{3} \\ 0 \end{pmatrix}$$

span $\mathrm{Col}(\mathbf{A})$.
(3) The left null space $\mathrm{Null}(\mathbf{A}^\mathsf{T})$ is spanned by

$$\begin{pmatrix} -1 \\ -1 \\ 4 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

(4) $\mathbf{A}$ has a left inverse.
(5) $\mathbf{A}$ has a right inverse.
(6) None of the above.

Problem 58. (2020 Spring POSTECH Applied Linear Algebra). For $a, b, c, d \in \mathbb{R}$, let

$$\mathbf{A} = \begin{pmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}.$$

Select all true statements.

(1) $\mathbf{A}$ is invertible if $abcd \neq 0$.
(2) $\mathbf{A}$ has a right inverse if $cd = 0$.
(3) $\mathbf{A}$ has both a right inverse and a left inverse if $ad - bc \neq 0$.
(4) $\mathbf{A}$ has a left inverse if $ab \neq 0$.
(5) None of the above.

Problem 59. (2020 Spring POSTECH Applied Linear Algebra). Let P_n be the vector space of real polynomials of degree at most n. Let E_n be the set of even polynomials in P_n, that is, polynomials p satisfying $p(x) = p(-x)$, and let O_n be the set of odd polynomials in P_n, that is, polynomials q satisfying $q(x) = -q(-x)$. Determine whether each of the following statements is true or false.

(1) E_n is a subspace of P_n.
(2) O_n is a subspace of P_n.
(3) The map
$$P_{n+1} \to P_n, \qquad f(x) \mapsto \frac{\mathrm{d}f}{\mathrm{d}x},$$
is linear.
(4) The map
$$P_n \to P_{2n+2}, \qquad f(x) \mapsto \int_1^{x^2} f(u)\ \mathrm{d}u,$$
is linear.
(5) $\dim P_n = n$.

Problem 60. (2020 Spring POSTECH Applied Linear Algebra). Let V be a one-dimensional vector space. Determine whether the following statement is true or false: if $f\colon V \to \mathbb{R}$ satisfies

$$f(a\mathbf{v}) = af(\mathbf{v})$$

for every $a \in \mathbb{R}$ and $\mathbf{v} \in V$, then f is linear.

Problem 61. (2020 Spring POSTECH Applied Linear Algebra). Let $V = M_3(\mathbb{R})$, the vector space of all 3×3 real matrices. Determine whether each of the following statements is true or false.

(1) $\dim V = 9$.
(2) The subset of invertible 3×3 matrices is a subspace of V.
(3) The subset of 3×3 matrices whose entries sum to zero is a subspace of V.
(4) The set of all linear transformations $V \to V$ is a vector space.
(5) The matrix representation of a linear transformation $V \to V$ is a 3×3 matrix.

Problem 62. (2020 Spring POSTECH Applied Linear Algebra). Let V be a vector space and let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$. Determine whether each of the following statements is true or false.

(1) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent, then $\mathbf{x}, \mathbf{y}$ are linearly independent.
(2) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly dependent, then

$$\mathrm{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \mathrm{Span}(\mathbf{x}, \mathbf{y}).$$

(3) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent, then

$$\dim \mathrm{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 3.$$

(4) If $\dim \mathrm{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 3$, then $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent.
(5) $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a basis for $\mathrm{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z})$.

Problem 63. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether each of the following statements is true or false.

(1) $\mathbf{Ax} = \mathbf{b}$ has a solution if and only if $\mathbf{b} \in \mathrm{Col}(\mathbf{A})$.
(2) $\mathbf{Ax} = \mathbf{b}$ has a unique solution if and only if $\mathrm{rk}(\mathbf{A}) = n$.
(3) If $\mathrm{rk}(\mathbf{A}) = m$, then $\mathbf{Ax} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$.
(4) If $m < n$, then $\mathbf{Ax} = \mathbf{b}$ has infinitely many solutions.
(5) If $m > n$, then there exists $\mathbf{b} \in \mathbb{R}^m$ for which $\mathbf{Ax} = \mathbf{b}$ has no solution.

Problem 64. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{U}$ be an echelon form of a matrix $\mathbf{A}$. Determine whether each of the following statements is true or false.

(1) $\mathrm{Row}(\mathbf{A}) = \mathrm{Row}(\mathbf{U})$.
(2) $\mathrm{Col}(\mathbf{A}) = \mathrm{Col}(\mathbf{U})$.
(3) $\mathrm{Null}(\mathbf{A}) = \mathrm{Null}(\mathbf{U})$.
(4) $\mathrm{Row}(\mathbf{A}) = \mathrm{Col}(\mathbf{A})$.
(5) $\mathrm{rk}(\mathbf{A}) = \mathrm{rk}(\mathbf{U})$.

Problem 65. (2020 Spring POSTECH Applied Linear Algebra). Let $U \subseteq \mathbb{R}^n$ be a subspace. Determine whether each of the following statements is true or false.

(1) Every $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{x} = \mathbf{u} + \mathbf{v}, \qquad \mathbf{u} \in U, \quad \mathbf{v} \in U^{\perp}.$$

(2) If $\mathbf{u} \in U$ and $\mathbf{v} \in U^{\perp}$, then $\mathbf{u}$ and $\mathbf{v}$ are linearly independent.
(3) $U \cap U^{\perp} = \{\mathbf{0}\}$.
(4) $\dim U = \dim U^{\perp}$.
(5) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfies $\mathrm{Col}(\mathbf{A}) = U$, then

$$\mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}$$

is the projection matrix onto U.

Problem 66. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be linearly independent vectors in $\mathbb{R}^5$. Applying the Gram–Schmidt process to $\mathbf{x}, \mathbf{y}, \mathbf{z}$ produces orthonormal vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \in \mathbb{R}^5$. Let $\mathbf{A}$ be the 5×3 matrix with columns $\mathbf{x}, \mathbf{y}, \mathbf{z}$, and let $\mathbf{Q}$ be the 5×3 matrix with columns $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Select all true statements.

(1) $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_3$.

(2) $\mathbf{Q}\mathbf{Q}^{\mathsf{T}} = \mathbf{I}_5$.
(3) $\mathbf{Q}$ has orthonormal rows.
(4) $\mathrm{Col}(\mathbf{A}) = \mathrm{Col}(\mathbf{Q})$.
(5)
$$\mathbf{u}_1\mathbf{u}_1^{\mathsf{T}} + \mathbf{u}_2\mathbf{u}_2^{\mathsf{T}} + \mathbf{u}_3\mathbf{u}_3^{\mathsf{T}}$$
is the projection matrix onto $\mathrm{Col}(\mathbf{Q})$.
(6) None of the above.

Problem 67. (2020 Spring POSTECH Applied Linear Algebra). Suppose that $\mathbf{Ax} = \mathbf{b}$ is inconsistent, and let $\mathbf{P}$ be the orthogonal projection onto $\mathrm{Col}(\mathbf{A})$. Determine whether each of the following statements is true or false.

(1) There is always a least-squares solution.
(2) The least-squares solution is unique.
(3) Every least-squares solution satisfies
$$\mathbf{A}^{\mathsf{T}}\mathbf{Ax} = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$
(4) Every solution of
$$\mathbf{Ax} = \mathbf{Pb}$$
is a least-squares solution.
(5)
$$\mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\mathbf{b}$$
is a least-squares solution.

Problem 68. (2020 Spring POSTECH Applied Linear Algebra). Let P_2 be the vector space of real polynomials of degree at most 2. Select the matrix representation of the linear transformation
$$T\colon P_2 \to \mathbb{R}^2, \qquad p(x) \mapsto (p(2), p'(2)),$$
with respect to the ordered bases $(1, x, x^2)$ of P_2 and $(\mathbf{e}_1, \mathbf{e}_2)$ of $\mathbb{R}^2$.

(1) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$
(2) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$
(3) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$
(4) $\begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \end{pmatrix}$
(5) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 4 & 4 \end{pmatrix}$

Problem 69. (2020 Spring POSTECH Applied Linear Algebra). Let V be a vector space. Determine whether the following statement is true or false: the matrix representation of the identity transformation
$$\mathrm{id}_V\colon V \to V, \qquad \mathbf{v} \mapsto \mathbf{v},$$
with respect to any choice of bases of V is an identity matrix.

Problem 70. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Determine whether the following statement is true or false:

$$\det(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A}) + \det(\mathbf{B}).$$

Problem 71. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false:

$$\det(2\mathbf{A}) = 2\det(\mathbf{A}).$$

Problem 72. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Determine whether the following statement is true or false:

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}).$$

Problem 73. (2020 Spring POSTECH Applied Linear Algebra). For

$$\det(\mathbf{A} - \lambda\mathbf{I}_3) = \det\begin{pmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{pmatrix},$$

what is the coefficient of λ^2?

(1) 2
(2) $a_{11} + a_{22} + a_{33}$
(3) $-a_{11}a_{22}a_{33}$
(4) $\det(\mathbf{A})$
(5) None of the above.

Problem 74. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an invertible matrix and let $\mathbf{C}$ be its cofactor matrix. Determine whether the following statement is true or false:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\mathbf{C}^{\mathsf{T}}.$$

Problem 75. (2020 Spring POSTECH Applied Linear Algebra). A parallelepiped in $\mathbb{R}^3$ has edge vectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}.$$

What is its volume?

(1) $\frac{10}{3}$
(2) 5
(3) 7.5
(4) 10
(5) None of the above.

Problem 76. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: the determinant of a permutation matrix is always 1.

Problem 77. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$. Determine whether the following statement is true or false: Cramer's rule gives a solution of $\mathbf{Ax} = \mathbf{b}$ for every such $\mathbf{A}$ and $\mathbf{b}$.

Problem 78. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ is diagonalizable.

Problem 79. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ has fewer than n distinct eigenvalues, then $\mathbf{A}$ is not diagonalizable.

Problem 80. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: eigenvectors of $\mathbf{A}$ associated with distinct eigenvalues are mutually orthogonal.

Problem 81. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ complex matrix. Determine whether the following statement is true or false: $\mathbf{A}$ is unitarily diagonalizable if and only if $\mathbb{C}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Problem 82. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: the solution of the initial value problem

$$\mathbf{u}'(t) = \mathbf{A}\mathbf{u}(t), \qquad \mathbf{u}(0) = \mathbf{u}_0,$$

is

$$\mathbf{u}(t) = e^{t\mathbf{A}}\mathbf{u}_0,$$

regardless of whether $\mathbf{A}$ is diagonalizable.

Problem 83. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every real symmetric matrix is orthogonally diagonalizable.

Problem 84. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every Hermitian matrix is unitarily diagonalizable.

Problem 85. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every orthogonally diagonalizable matrix is real symmetric.

Problem 86. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices. Determine whether the following statement is true or false: if $\mathbf{A}$ and $\mathbf{B}$ are congruent, then they have the same eigenvalues.

Problem 87. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices. Determine whether the following statement is true or false: if $\mathbf{A}$ and $\mathbf{B}$ are congruent, then they have the same number of positive eigenvalues.

Problem 88. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every real symmetric matrix is congruent to a diagonal matrix.

Problem 89. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a real symmetric matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ is positive definite, then every leading principal submatrix of $\mathbf{A}$ has positive determinant.

Problem 90. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Determine whether the following statement is true or false: every real matrix has a singular value decomposition.

Problem 91. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Determine whether the following statement is true or false: the singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of $\mathbf{A}^\mathsf{T}\mathbf{A}$.

Problem 92. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Determine whether the following statement is true or false: if $\mathbf{A}$ has a left inverse, then its pseudoinverse is a left inverse of $\mathbf{A}$.

Problem 93. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every matrix has a pseudoinverse.

Problem 94. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Determine whether the following statement is true or false: $\mathbf{A}^\mathsf{T}\mathbf{A}$ and $\mathbf{A}\mathbf{A}^\mathsf{T}$ have the same eigenvalues.

Problem 95. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 2×2 upper triangular matrix with diagonal entries 1 and 2. Determine whether the following statement is true or false: $\mathbf{A}$ is diagonalizable.

Problem 96. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following matrix is diagonalizable:

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

Problem 97. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^{1000} = \begin{pmatrix} 1 & 0 \\ 0 & 2^{1000} \end{pmatrix}.$$

Problem 98. (2020 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}^{-1}.$$

Determine whether the following statement is true or false: the vectors

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

are eigenvectors of $\mathbf{A}$.

Problem 99. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 2×2 matrix whose eigenvalues are 1 and 2. Determine whether the following statement is true or false: $\mathbf{A}$ is invertible.

Problem 100. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 2×2 matrix whose only eigenvalue is 2. Determine whether the following statement is true or false: $\mathbf{A}$ is not invertible.

Problem 101. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 2×2 matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ is not diagonalizable, then it has only one eigenvalue, counted with algebraic multiplicity two.

Problem 102. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 2×2 matrix with repeated eigenvalue 3. Determine whether the following statement is true or false: $\mathbf{A}$ is invertible.

Problem 103. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 matrices. Determine whether the following statement is true or false:

$$e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}} e^{\mathbf{B}}.$$

Problem 104. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $n \times n$ complex matrix. Determine whether the following statement is true or false: either

$$\det(\mathbf{A}) = \det(\mathbf{A}^{\mathsf{H}})$$

or

$$\det(\mathbf{A}) = -\det(\mathbf{A}^{\mathsf{H}}).$$

Problem 105. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ orthogonal matrices. Determine whether the following statement is true or false: $\mathbf{AB}$ is orthogonal.

Problem 106. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 5×5 matrix with rank 4. Determine whether the following statement is true or false:

$$\det(\mathbf{A}) = 0.$$

Problem 107. (2020 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

Determine whether the following statement is true or false: $\mathbf{A} - \alpha \mathbf{I}_2$ is singular only when $\alpha = 2$ or $\alpha = 5$.

Problem 108. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every full-rank 2×2 matrix is diagonalizable.

Problem 109. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: it is possible to choose $x \in \mathbb{R}$ such that $\det(\mathbf{A}) \neq 0$ for

$$\mathbf{A} = \begin{pmatrix} x & 0 & 0 \\ x & 0 & 0 \\ x & x & x \end{pmatrix}.$$

Problem 110. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: the only possible eigenvalues of a real symmetric orthogonal matrix are 1 and -1.

Problem 111. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: similar matrices have the same eigenvalues, counting algebraic multiplicity.

Problem 112. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: if two $n \times n$ matrices have the same eigenvalues, counting algebraic multiplicity, then they are similar.

Problem 113. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a symmetric matrix. Determine whether the following statement is true or false: $\mathbf{A}$ is positive definite if and only if there exists a matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{R}^{\mathsf{T}}\mathbf{R}.$$

Problem 114. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 4×4 real symmetric matrix with $\det(\mathbf{A}) = 30$. Suppose that $\mathbf{A}$ is not positive definite, all eigenvalues of $\mathbf{A}$ are integers, and two eigenvalues are 2 and 3. What are the other two eigenvalues?

(1) $1, 5$
(2) $-1, -5$
(3) $1, -5$
(4) $2, 3$
(5) None of the above.

Problem 115. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every symmetric matrix with positive determinant is positive definite.

Problem 116. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a square matrix. Determine whether the following statement is true or false: if every cofactor of $\mathbf{A}$ is zero, then $\mathbf{A}$ is not invertible.

Problem 117. (2020 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a 3×3 matrix with exactly two linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, corresponding to eigenvalues λ_1, λ_2, respectively.

(a) Determine whether the following statement is true or false: λ_1 and λ_2 must be distinct.
(b) Select a Jordan form to which $\mathbf{A}$ is similar.

(1) $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(2) $\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(3) $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(4) $\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(5) None of the above.

Problem 118. (2020 Spring POSTECH Applied Linear Algebra). Determine whether the following statement is true or false: every complex square matrix has at least one eigenvector.

Problem 119. (2020 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(a) Solve $\mathbf{Ax} = \mathbf{b}$.

(b) Find an LDU decomposition

$$\mathbf{A} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

(c) Find $\mathbf{A}^{-1}$.

Problem 120. (2020 Fall POSTECH Applied Linear Algebra). For

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix},$$

find bases for the row space $\mathrm{Row}(\mathbf{A})$, equivalently $\mathrm{Col}(\mathbf{A}^\mathsf{T})$, and the null space $\mathrm{Null}(\mathbf{A})$.

Problem 121. (2020 Fall POSTECH Applied Linear Algebra). Let V be the subspace of $\mathbb{R}^4$ spanned by the orthonormal vectors

$$\begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}, \qquad \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \qquad \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Find the orthogonal projection of

$$\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$$

onto V.

Problem 122. (2020 Fall POSTECH Applied Linear Algebra). Let V be a subspace of $\mathbb{R}^n$ and let $W = V^\perp$. Show that

$$V + W = \mathbb{R}^n.$$

Problem 123. (2020 Fall POSTECH Applied Linear Algebra). Let $\mathfrak{sl}_2$ be the set of 2×2 real matrices whose diagonal entries sum to zero.

(a) Show that $\mathfrak{sl}_2$ is a vector space.
(b) Find a basis for $\mathfrak{sl}_2$, and prove that your chosen matrices form a basis.
(c) Find $\dim \mathfrak{sl}_2$.

Problem 124. (2020 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{H} = \mathbf{I}_n - 2\mathbf{u}\mathbf{u}^\mathsf{T},$$

where $\mathbf{u} \in \mathbb{R}^n$ is a unit vector.

(a) Show that

$$\mathbf{H}\mathbf{u} = -\mathbf{u}$$

and

$$\mathbf{H}\mathbf{w} = \mathbf{w}$$

for every $\mathbf{w} \perp \mathbf{u}$.
(b) Show that $\|\mathbf{H}\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x}$ of compatible dimension.
(c) For $n = 3$, let

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} \sqrt{2} \\ 0 \\ 0 \end{pmatrix}.$$

Find a unit vector $\mathbf{u}$ such that $\mathbf{x}$ solves

$$\mathbf{H}\mathbf{x} = \mathbf{b}.$$

Problem 125. (2020 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 & -2 \\ -2 & -1 & 2 \\ 3 & -1 & -5 \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$.
(b) Find the eigenvectors of $\mathbf{A}$.
(c) Solve

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t), \qquad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

and express each component of $\mathbf{y}(t)$ explicitly.

Problem 126. (2020 Fall POSTECH Applied Linear Algebra). Let

$$V = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \subseteq \mathbb{R}^n$$

be a set of nonzero mutually orthogonal vectors.

(a) Show that V is a basis for $\mathbb{R}^n$.

(b) Let $\mathbf{A}$ be the $n \times n$ matrix whose columns are $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Find $\det(\mathbf{A})$ in terms of $\mathbf{v}_1, \ldots, \mathbf{v}_n$.

(c) Let $\mathbf{B}$ be any $n \times n$ matrix. Show that

$$|\det(\mathbf{B})| = \mathrm{vol}(P),$$

where P is the parallelepiped spanned by the columns of $\mathbf{B}$.

Problem 127. (2020 Fall POSTECH Applied Linear Algebra).

(a) Find a 6×6 matrix $\mathbf{A}$ satisfying

$$\det(\mathbf{A} - \lambda \mathbf{I}_6) = (\lambda - 2)(\lambda - 3)^2(\lambda - 4)^3,$$

and

$$\dim E_2(\mathbf{A}) = 1, \qquad \dim E_3(\mathbf{A}) = 1, \qquad \dim E_4(\mathbf{A}) = 2.$$

(b) Compute $e^{t\mathbf{A}}$ for the matrix $\mathbf{A}$ found in part (a).

Problem 128. (2020 Fall POSTECH Applied Linear Algebra). Consider the Markov matrix

$$\mathbf{A} = \begin{pmatrix} 0.5 & 0.5 & 0.25 \\ 0 & 0.5 & 0.25 \\ 0.5 & 0 & 0.5 \end{pmatrix}.$$

(a) Determine whether the sequence

$$\mathbf{A}^k \begin{pmatrix} 0.25 \\ 0.25 \\ 0.5 \end{pmatrix}$$

converges as $k \to \infty$.

(b) Give an example of a Markov matrix $\mathbf{M}$ and a column vector $\mathbf{u}_0$ for which the sequence $\mathbf{M}^k \mathbf{u}_0$ does not converge as $k \to \infty$.

Problem 129. (2020 Fall POSTECH Applied Linear Algebra). For

$$f(x, y, z) = 2x^2 + 5y^2 + 8z^2 + 4xy + 6yz,$$

let $\mathbf{A}$ be a symmetric matrix satisfying

$$f(x, y, z) = \mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{x}, \qquad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Find $\mathbf{A}$.

(b) Show that $\mathbf{A}$ is positive definite using determinants.

(c) Show that $\mathbf{A}$ is positive definite using pivots.

(d) Show that $\mathbf{A}$ is positive definite using eigenvalues. *Hint.* 10 is an eigenvalue.

Problem 130. (2020 Fall POSTECH Applied Linear Algebra). Suppose a vector $\mathbf{p}_0 \in \mathbb{R}^3$ rotates at a constant angular rate of 1 rad/s about the unit axis

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}, \qquad \|\mathbf{w}\| = 1.$$

Let $\mathbf{p}(t)$ denote its position at time t. Its velocity is

$$\mathbf{p}'(t) = \mathbf{w} \times \mathbf{p}(t).$$

Since $\mathbf{w} \times \mathbf{p}(t) = \mathbf{\Omega}\mathbf{p}(t)$, where

$$\mathbf{\Omega} = \begin{pmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{pmatrix},$$

we have

$$\mathbf{p}'(t) = \mathbf{\Omega}\mathbf{p}(t), \qquad \mathbf{p}(0) = \mathbf{p}_0.$$

(a) Find the eigenvalues of $\mathbf{\Omega}$.

(b) Show that

$$e^{t\mathbf{\Omega}} = \mathbf{I}_3 + (\sin t)\mathbf{\Omega} + (1 - \cos t)\mathbf{\Omega}^2.$$

Hint. Use the power-series expansions of $\sin t$ and $\cos t$.

(c) Show that the position vector preserves its length:

$$\|\mathbf{p}(t)\| = \|\mathbf{p}_0\|$$

for all t.

(d) Suppose

$$\mathbf{w} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{p}_0 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Find $\mathbf{p}(\pi/2)$.

Problem 131. (2020 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \mathbf{I}_n - \mathbf{u}\mathbf{v}^{\mathsf{T}}, \qquad \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

(a) Show that if $\mathbf{A}$ is invertible, then its inverse has the form

$$\mathbf{A}^{-1} = \mathbf{I}_n + \alpha\mathbf{u}\mathbf{v}^{\mathsf{T}}$$

for some scalar α, and find α.

(b) Determine for which $\mathbf{u}$ and $\mathbf{v}$ the matrix $\mathbf{A}$ is not invertible.

(c) If $\mathbf{A}$ is not invertible, find $\mathrm{Null}(\mathbf{A})$.

Problem 132. (2020 Fall POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

(a) Find a diagonal matrix $\mathbf{\Lambda}$ and an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}.$$

(b) Compute $\mathbf{A}^{2020}$, describing each entry exactly.

(c) Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation represented by $\mathbf{A}$. Find an orthonormal basis of $\mathbb{R}^3$ with respect to which the matrix representation of T is diagonal.

Problem 133. (2021 Spring POSTECH Applied Linear Algebra). Find an *LDU* decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 4 & 4 \end{pmatrix} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

Problem 134. (2021 Spring POSTECH Applied Linear Algebra). Find an *LDU* decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 1 & 1 & 4 \end{pmatrix} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

Problem 135. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}.$$

Find a *QR* decomposition $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is invertible and upper triangular.

Problem 136. (2021 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and assume that $\mathbf{A}^\mathsf{T}\mathbf{A}$ is invertible. Define

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

For $\mathbf{b} \in \mathbb{R}^m$, show that $\mathbf{b} - \mathbf{Pb}$ is orthogonal to every vector in $\mathrm{Col}(\mathbf{A})$.

Problem 137. (2021 Spring POSTECH Applied Linear Algebra). Determine whether each of the following statements is true or false for every matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$. You need not justify your answers.

(1) $\mathbf{A}$ and $-\mathbf{A}$ have the same reduced row echelon form.
(2) If $\mathbf{A}$ is invertible, then $\mathbf{A}^\mathsf{T}$ is invertible.
(3) If $\mathbf{Ax} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$, then $n \geq m$.
(4) If $\mathbf{Ax} = \mathbf{b}$ has a unique solution for some $\mathbf{b} \in \mathbb{R}^m$, then $n \leq m$.
(5) If the columns of a square matrix $\mathbf{A}$ are orthonormal, then its rows are orthonormal.
(6) If $\mathbf{Ax} = \mathbf{b}$ has infinitely many solutions for some $\mathbf{b} \in \mathbb{R}^m$, then $\mathbf{Ax} = \mathbf{0}$ has a nonzero solution.
(7) $\mathrm{rk}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \mathrm{rk}(\mathbf{A})$.

Problem 138. (2021 Spring POSTECH Applied Linear Algebra). Find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 3 & 4 & 1 & 0 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

Problem 139. (2021 Spring POSTECH Applied Linear Algebra). Let P_4 be the vector space of real polynomials of degree at most 4, and define the linear map

$$T\colon P_4 \to P_4, \qquad T(p(t)) = t^2 \frac{\mathrm{d}^2 p(t)}{\mathrm{d}t^2}.$$

(a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$(1, t, t^2, t^3, t^4).$$

(b) Find the matrix $\mathbf{B}$ representing T with respect to the ordered basis

$$(1, t+1, t^2+1, t^3+1, t^4+1).$$

Problem 140. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 1 & 1 & 1 \\ 1 & 4 & 2 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

(a) Find a basis for $\mathrm{Col}(\mathbf{A})$.
(b) Find a basis for $\mathrm{Col}(\mathbf{A})^{\perp}$.
(c) Find an orthonormal basis for $\mathrm{Null}(\mathbf{A})$.

Problem 141. (2021 Spring POSTECH Applied Linear Algebra). State precisely the definition of each of the following concepts.

(1) Linear independence.
(2) Orthogonal matrix.
(3) Vector space.
(4) Projection matrix.
(5) Linear transformation.

Problem 142. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 2 & 4 & 3 \end{pmatrix}.$$

(a) Find $\mathbf{A}^{-1}$. Write your answer as a single explicit 3×3 matrix.
(b) Find an LDU decomposition of $\mathbf{A}$.

Problem 143. (2021 Spring POSTECH Applied Linear Algebra). Let

$$V = \mathrm{Span}\left(\begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right) \subseteq \mathbb{R}^5.$$

(a) Find $V^{\perp}$.
(b) Find all $\mathbf{x} \in \mathbb{R}^5$ satisfying $\mathbf{A}\mathbf{x} = \mathbf{b}$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 2 & -2 \\ 2 & 2 & 1 & 1 & 0 \\ 3 & 0 & 0 & 1 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Problem 144. (2021 Spring POSTECH Applied Linear Algebra). Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be reflection across the plane

$$z = 2y.$$

Find the matrix representing T with respect to the standard basis. Write your answer as a single explicit 3×3 matrix.

Problem 145. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & -1 & -1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Find the minimum value of

$$\|\mathbf{Ax} - \mathbf{b}\|^2.$$

Problem 146. (2021 Spring POSTECH Applied Linear Algebra). Let V be a vector space. Suppose that

$$\{\mathbf{v}_1, \dots, \mathbf{v}_n\} \qquad \text{and} \qquad \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$$

are two bases of V. Show that $m = n$.

Problem 147. (2021 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Show that if either $\mathbf{A}$ or $\mathbf{B}$ is not invertible, then $\mathbf{AB}$ is not invertible.

Problem 148. (2021 Spring POSTECH Applied Linear Algebra). Let $\mathbf{Q}$ be an $n \times n$ matrix. Show that if

$$\|\mathbf{Qx}\| = \|\mathbf{x}\|$$

for every $\mathbf{x} \in \mathbb{R}^n$, then the rows of $\mathbf{Q}$ are orthonormal.

Problem 149. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{pmatrix}.$$

(a) Find an echelon form of $\mathbf{A}$.
(b) Find $\mathrm{rk}(\mathbf{A})$.
(c) Find a basis for $\mathrm{Col}(\mathbf{A})$.
(d) Find a basis for $\mathrm{Col}(\mathbf{A}^\mathsf{T})$.
(e) Find an LDU factorization of $\mathbf{A}$.
(f) Solve $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{b} = \begin{pmatrix} 16 \\ 20 \\ 12 \end{pmatrix}.$$

Problem 150. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 13 \end{pmatrix}.$$

Show that the columns of $\mathbf{A}$ are linearly independent.

Problem 151. (2021 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Using the Gauss–Jordan method, find $\mathbf{A}^{-1}$.

Problem 152. (2021 Spring POSTECH Applied Linear Algebra). Let $V, W \subseteq \mathbb{R}^n$ be subspaces.

(a) Prove that $V + W$ is a subspace of $\mathbb{R}^n$.

(b) Prove that $V \cap W$ is a subspace of $\mathbb{R}^n$.

(c) Prove that

$$\dim(V + W) + \dim(V \cap W) = \dim V + \dim W.$$

Problem 153. (2023 Spring POSTECH Applied Linear Algebra). Determine whether each of the following statements is true or false. You do not need to justify your answers.

(1) Let $\mathbf{A}$ and $\mathbf{B}$ be matrices. If both $\mathbf{AB}$ and $\mathbf{BA}$ are defined, then both products are square matrices.

(2) Let $\mathbf{A}$ be a square matrix. Suppose that

$$\mathbf{A} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with all diagonal entries equal to 1, $\mathbf{U}$ is upper triangular with all diagonal entries equal to 1, and $\mathbf{D}$ is diagonal with no zero diagonal entries. Then such a decomposition is unique.

(3) If $\mathbf{A}$ is a square matrix and $\mathbf{A}^2$ is nonsingular, then $\mathbf{A}$ is nonsingular.

(4) The set of all $n \times n$ skew-symmetric matrices,

$$\mathbf{A}^{\mathsf{T}} = -\mathbf{A},$$

forms a subspace of the vector space of $n \times n$ matrices.

(5) Every subspace of $\mathbb{R}^4$ is the null space of some matrix.

(6) Let $H_\theta \colon \mathbb{R}^2 \to \mathbb{R}^2$ denote reflection across the line through the origin making angle θ with the positive x-axis. Then $H_{\theta_1} \circ H_{\theta_2}$ is counterclockwise rotation by $2(\theta_1 - \theta_2)$.

(7) There exists a nonsquare matrix that has both a left inverse and a right inverse, possibly different from each other.

(8) Let V and W be finite-dimensional vector spaces. If a linear transformation $T \colon V \to W$ is injective, then

$$\dim V \leq \dim W.$$

(9) Let $T \colon \mathbb{R}^n \to \mathbb{R}^m$ be linear. If the values $T(\mathbf{v})$ are known for n distinct nonzero vectors $\mathbf{v} \in \mathbb{R}^n$, then $T(\mathbf{v})$ is determined for every $\mathbf{v} \in \mathbb{R}^n$.

(10) For a square matrix $\mathbf{A}$, if the columns of $\mathbf{A}$ are linearly independent, then the rows of $\mathbf{A}$ are linearly independent.

Problem 154. (2023 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A} = (a_{ij})_{n\times n}$ be the matrix defined by

$$a_{ij} = \begin{cases} 1, & j - i = 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Evaluate the sum of all entries of $\mathbf{A}$,

$$\sum_{1\le i,j\le n} a_{ij}.$$

(b) Let $\mathbf{B} = (b_{ij})$ be

$$\mathbf{B} = \sum_{k=0}^{n-1} \mathbf{A}^k = \mathbf{I}_n + \mathbf{A} + \mathbf{A}^2 + \cdots + \mathbf{A}^{n-1}.$$

Evaluate the sum of all entries of $\mathbf{B}$,

$$\sum_{1\le i,j\le n} b_{ij}.$$

Problem 155. (2023 Spring POSTECH Applied Linear Algebra). Consider

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & -2 \\ 0 & a & 4 \end{pmatrix}.$$

(a) Find all $a \in \mathbb{R}$ for which $\mathbf{A}$ is invertible.

(b) Using the Gauss–Jordan method, find $\mathbf{A}^{-1}$ when $a = 2$.

Problem 156. (2023 Spring POSTECH Applied Linear Algebra). Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 2 & 4 & 8 & 6 \\ -1 & -2 & 1 & 2 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

In solving $\mathbf{Ax} = \mathbf{b}$, apply Gaussian elimination to obtain

$$\mathbf{Ux} = \mathbf{c},$$

and then reduce further to

$$\mathbf{Rx} = \mathbf{d},$$

where $\mathbf{U}$ is the echelon matrix obtained from $\mathbf{A}$ and $\mathbf{R}$ is the reduced row echelon form of $\mathbf{A}$. Compute $\mathbf{c}$ and $\mathbf{d}$ in terms of b_1, b_2, b_3.

Problem 157. (2023 Spring POSTECH Applied Linear Algebra). Let $V = M_2(\mathbb{R})$ be the vector space of 2×2 real matrices, and let P_3 be the vector space of real polynomials of degree at most 3. Consider the linear map $T\colon P_3 \to V$ defined by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = \begin{pmatrix} a_0 + a_1 & -a_0 - a_2 \\ -a_1 + a_3 & a_2 - a_3 \end{pmatrix}.$$

(a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$(1, x, x^2, x^3)$$

of P_3 and the ordered basis $(\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{E}_4)$ of V, where

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{E}_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

(b) Find $\operatorname{rk}(\mathbf{A})$ and $\dim \operatorname{Null}(\mathbf{A})$.

Problem 158. (2023 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$$

Decompose

$$\mathbf{b} = \mathbf{b}_c + \mathbf{b}_n,$$

where $\mathbf{b}_c \in \mathrm{Col}(\mathbf{A})$ and $\mathbf{b}_n \in \mathrm{Null}(\mathbf{A}^\mathsf{T})$, in the following two ways.

(a) Solve the normal equation

$$\mathbf{A}^\mathsf{T}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b}$$

and compute $\mathbf{b}_c = \mathbf{A}\hat{\mathbf{x}}$.

(b) Find an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2\}$ for $\mathrm{Col}(\mathbf{A})$ and express $\mathbf{b}_c$ as a linear combination of $\mathbf{q}_1$ and $\mathbf{q}_2$.

Problem 159. (2023 Spring POSTECH Applied Linear Algebra). Suppose that data are expected to conform to the quadratic relation

$$y(t) = A + Bt + Ct^2.$$

The observed data are

t	-2	-1	0	1	2
$y(t)$	-4	1	3	4	6

Find the parameters A, B, C that minimize the sum of squared errors.

Problem 160. (2023 Spring POSTECH Applied Linear Algebra). Let V be a subspace of $\mathbb{R}^d$. Prove that if

$$\{\mathbf{v}_1, \ldots, \mathbf{v}_m\} \qquad \text{and} \qquad \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$$

are both bases for V, then $m = n$.

Problem 161. (2023 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be an $m \times n$ real matrix. Prove that

$$\mathrm{rk}(\mathbf{A}) = \mathrm{rk}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \mathrm{rk}(\mathbf{A}\mathbf{A}^\mathsf{T}).$$

Problem 162. (2023 Spring POSTECH Applied Linear Algebra). Determine whether each of the following statements is true or false. You do not need to justify your answers.

(1) There exist 3×3 matrices $\mathbf{A}$ and $\mathbf{B}$ such that

$$\mathbf{AB} - \mathbf{BA} = \mathbf{I}_3.$$

(2) If $\mathbf{A}$ and $\mathbf{B}$ are simultaneously diagonalizable, and $\mathbf{B}$ and $\mathbf{C}$ are simultaneously diagonalizable, then $\mathbf{A}$ and $\mathbf{C}$ are simultaneously diagonalizable.

(3) There exists a real 2×2 matrix $\mathbf{A} \neq \mathbf{I}_2$ such that

$$\mathbf{A}^3 = \mathbf{I}_2.$$

(4) If a 3×3 matrix $\mathbf{A}$ has eigenvalues $1, 2, 3$, then

$$(\mathbf{A} - \mathbf{I}_3)(\mathbf{A} - 2\mathbf{I}_3)(\mathbf{A} - 3\mathbf{I}_3)$$

cannot have a nonzero eigenvalue.

(5) If a real matrix $\mathbf{A}$ is normal, then

$$(\mathbf{A}^k)^+ = (\mathbf{A}^+)^k$$

for every positive integer k.

(6) If $\det(\mathbf{A}) > 0$ and $\det(\mathbf{B}) > 0$, then

$$\det(\mathbf{A} + \mathbf{B}) > 0.$$

(7) If $\mathbf{A} = (a_{ij})$ is an $n \times n$ matrix satisfying

$$\sum_{i=1}^{n} a_{ij} = 5$$

for every j, then 5 is an eigenvalue of $\mathbf{A}$.

(8) If $\mathbf{A}$ is an $n \times n$ symmetric matrix, then $e^{\mathbf{A}}$ is positive definite.

(9) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. If λ is an eigenvalue of $\mathbf{A}$ and μ is an eigenvalue of $\mathbf{B}$, then $\lambda + \mu$ is an eigenvalue of $\mathbf{A} + \mathbf{B}$.

(10) If a real symmetric matrix $\mathbf{A}$ is positive definite, then it is nonsingular.

Problem 163. (2023 Spring POSTECH Applied Linear Algebra).

(a) Find the volume of the parallelepiped determined by the edge vectors from $(0, 0, 0)$ to

$$(-1, 2, 2), \qquad (2, -1, 2), \qquad (2, 2, -1).$$

(b) Compute

$$\det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Problem 164. (2023 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be $n \times n$ matrices and let $\mathbf{0}$ be the $n \times n$ zero matrix. Show that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{C}).$$

Hint. One possible approach is to use the Leibniz formula for the determinant.

Problem 165. (2023 Spring POSTECH Applied Linear Algebra). Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be sequences satisfying

$$a_{n+1} = 3a_n + 2b_n, \qquad b_{n+1} = a_n + 4b_n, \qquad a_0 = 1, \qquad b_0 = 0.$$

Evaluate

$$\lim_{n \to \infty} \frac{a_n - b_n}{a_n + b_n}.$$

Problem 166. (2023 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find two matrices among $\mathbf{A}, \mathbf{B}, \mathbf{C}$ that are similar to each other. Justify your answer.

Problem 167. (2023 Spring POSTECH Applied Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

(a) Find the singular value decomposition of $\mathbf{A}$.
(b) Find the pseudoinverse $\mathbf{A}^+$ of $\mathbf{A}$.

Problem 168. (2023 Spring POSTECH Applied Linear Algebra).

(a) Find all $\alpha \in \mathbb{R}$ such that

$$\mathbf{A} = \begin{pmatrix} 4 & \alpha \\ \alpha & 5 \end{pmatrix}$$

is positive definite.

(b) For a real number α such that $\mathbf{A}$ is positive definite, let $d_1(\alpha)$ and $d_2(\alpha)$ be the distances from the origin to, respectively, the closest and farthest points on

$$4x^2 + 2\alpha xy + 5y^2 = 1.$$

Find the minimum value of

$$f(\alpha) = d_1(\alpha)^2 + d_2(\alpha)^2.$$

Problem 169. (2023 Spring POSTECH Applied Linear Algebra). Let $\mathbf{A}$ be a square matrix with real entries. Show that

$$\operatorname{tr}(\mathbf{A}^\mathsf{T}\mathbf{A}) = 0$$

if and only if $\mathbf{A} = \mathbf{0}$.

Problem 170. (2023 Spring POSTECH Applied Linear Algebra). For an $n \times n$ matrix $\mathbf{A}$, show that

$$\det(e^{\mathbf{A}}) = e^{\operatorname{tr}(\mathbf{A})}.$$

Hint. By Schur's triangularization theorem over $\mathbb{C}$, there exists an invertible matrix $\mathbf{M}$ such that $\mathbf{M}^{-1}\mathbf{A}\mathbf{M}$ is upper triangular.

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Problem 171. (2022 Spring KAIST Introduction to Linear Algebra). Describe all possible 2×4 reduced row echelon forms whose rows are linearly independent.

Problem 172. (2022 Spring KAIST Introduction to Linear Algebra).

(a) Let

$$\mathbf{A}(\theta) = \begin{pmatrix} \cos\theta & \csc\theta & 0 & -\sin\theta \\ \cos^2\theta & \cot\theta & 2.78 & \sin^2\theta \\ 0 & 3.14 & 0 & 0 \\ \sin\theta & \tan\theta & 0 & \cos\theta \end{pmatrix}.$$

Compute

$$\det(\mathbf{A}(1.76\pi)) - \det(\mathbf{A}(2\pi/3)).$$

(b) Suppose $\mathbf{A} \in \mathbb{R}^{3\times 3}$ and $\det(\mathbf{A}) = 2$. Compute

$$\det(\mathbf{A}^{-1}) + \det(3\mathbf{A}^{-1}) + \det((5\mathbf{A})^{-1}) + \det(\mathbf{A}^\mathsf{T}).$$

Problem 173. (2022 Spring KAIST Introduction to Linear Algebra). There are three main colleges of science and engineering at KAIST: the College of Natural Sciences (NS), the College of Life Science and Bioengineering (LSB), and the College of Engineering (E). Suppose students pursuing transdisciplinary study choose one of the following three course sets:

(1) Type I: 30 credits from NS, 21 credits from LSB, and 27 credits from E;
(2) Type II: 33 credits from NS, 21 credits from LSB, and 30 credits from E;
(3) Type III: 30 credits from NS, 15 credits from LSB, and 39 credits from E.

Among graduates in 2022 who chose one of these three sets, the total numbers of credits taken were 4950 from NS, 2880 from LSB, and 5430 from E. How many students chose Type I?

Problem 174. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{pmatrix}.$$

Find $\mathbf{A}^{-1}$.

Problem 175. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have linearly independent columns. Show that if $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$ are linearly independent, then

$$\mathbf{A}\mathbf{v}_1, \dots, \mathbf{A}\mathbf{v}_p$$

are linearly independent.

Problem 176. (2022 Spring KAIST Introduction to Linear Algebra). Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator satisfying

$$\|T(\mathbf{x})\| = \|\mathbf{x}\| \qquad \text{for every } \mathbf{x} \in \mathbb{R}^n.$$

Prove that

$$T(\mathbf{x}) \cdot T(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y} \qquad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

Problem 177. (2022 Spring KAIST Introduction to Linear Algebra). Show that

$$\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

represents a rotation R_θ about a line ℓ through the origin, followed by a reflection H_ℓ about the plane through the origin perpendicular to ℓ. Find the standard matrices of R_θ and H_ℓ.

Problem 178. (2022 Spring KAIST Introduction to Linear Algebra). For $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, one has

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

Prove or disprove the following statement: for $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$,

$$\operatorname{tr}(\mathbf{ABC}) = \operatorname{tr}(\mathbf{BAC}).$$

Problem 179. (2022 Spring KAIST Introduction to Linear Algebra). Suppose $p(x)$ and $q(x)$ are polynomials with real coefficients, and let $r(x) = p(x)q(x)$. Show that for every $n \times n$ matrix $\mathbf{A}$,

$$r(\mathbf{A}) = p(\mathbf{A})q(\mathbf{A}).$$

Problem 180. (2022 Spring KAIST Introduction to Linear Algebra).

(a) Find all 2×2 orthogonal matrices that do not admit an LU decomposition.

(b) For each such orthogonal matrix, find a PLU decomposition.

Problem 181. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n\times n}$ be triangular, with pairwise distinct diagonal entries:

$$a_{ii} \neq a_{jj} \qquad \text{whenever } i \neq j.$$

Find all eigenvalues of $\mathbf{A}$ and determine the algebraic multiplicity of each eigenvalue.

Problem 182. (2022 Spring KAIST Introduction to Linear Algebra). Prove or disprove the following statement. For a consistent linear system

$$\mathbf{Ax} = \mathbf{b}$$

with n unknowns and m equations, the solution set has the form

$$\mathbf{x}_0 + \mathrm{Null}(\mathbf{A}) = \{\mathbf{x}_0 + \mathbf{v} \mid \mathbf{v} \in \mathrm{Null}(\mathbf{A})\}$$

for some solution $\mathbf{x}_0$.

Problem 183. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be nilpotent; that is, $\mathbf{A}^k = \mathbf{0}$ for some positive integer k. Show that the matrix transformation

$$T_{\mathbf{A}} \colon \mathbb{R}^n \to \mathbb{R}^n, \qquad T_{\mathbf{A}}(\mathbf{x}) = \mathbf{Ax},$$

is not injective.

Problem 184. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{3\times 3}$ be skew-symmetric, so that

$$\mathbf{A}^{\mathsf{T}} = -\mathbf{A}.$$

(a) Find the characteristic polynomial of $\mathbf{A}$ and all real eigenvalues of $\mathbf{A}$.

(b) Find the full eigenspace corresponding to each real eigenvalue.

Problem 185. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \frac{1}{6}\begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

Draw $\mathrm{Col}(\mathbf{A})$ and $\mathrm{Null}(\mathbf{A})$ in $\mathbb{R}^3$. It is known that $\mathbf{A}$ is the standard matrix of the orthogonal projection onto $\mathrm{Col}(\mathbf{A})$. For n randomly generated points, transform them as specified in the MATLAB code below and scatter-plot the original and transformed points. Fill in blanks (1)–(10).

```
%%% plot the column space and the nullspace of A %%%
A = 1/6 * [1, 1, 2; 1, 1, 2; 2, 2, 4];
x = ___(1)___; % generate a vector that contains
    % 150 equally spaced points, from -1 to 1.
y = x;
[X, Y] = ___(2)___(x, y); % construct a grid over the domain
a = [___(3)___]; % choose a column vector "a" in the column space of A
plot3(x, ___(4)___, ___(5)___) % draw the column space of A
hold on;
Z = ___(6)___; % compute the z-components of the nullspace (plane) of A
___(7)___; % draw the surface of the nullspace of A
hold on; % keep using the same figure window for future plots
%%% scatter plot n randomly generated points and their transformed points %%%
n = 5;
b = -1 + 2*rand(3,n); % generate n random points in [-1,1]^3
p = zeros(3,n);
for i=1:n
    % if norm(b(:,i)) is greater than or equal to 1/2,
    % project b(:,i) onto the column space of A;
    % otherwise stretch its entries by 1, 0.5, and 2, respectively
    ___(8)___
        p(:,i) = ___(9)___;
    else
        p(:,i) = ___(10)___;
    end
end
scatter3(b(1,:),b(2,:),b(3,:),'*');
scatter3(p(1,:),p(2,:),p(3,:));
axis equal;
```

Problem 186. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n\times n}$. Determine whether each of the following statements is true or false, and justify your answer.

(a) Elementary row operations do not change $\mathrm{Row}(\mathbf{A})$.

(b) Elementary row operations do not change $\mathrm{Col}(\mathbf{A})$.

(c) Elementary row operations do not change $\dim \mathrm{Null}(\mathbf{A}^{\mathsf{T}})$.

(d) The nonzero rows of any row echelon form $\mathbf{R}$ of $\mathbf{A}$ form a basis for $\mathrm{Row}(\mathbf{A})$, even when $\mathbf{R}$ is not in reduced row echelon form.

(e) The nonzero columns of any row echelon form $\mathbf{R}$ of $\mathbf{A}$ form a basis for $\mathrm{Col}(\mathbf{A})$, even when $\mathbf{R}$ is not in reduced row echelon form.

Problem 187. (2022 Spring KAIST Introduction to Linear Algebra). Given $\mathbf{A} \in \mathbb{R}^{m\times n}$, let $\mathcal{B}$ be a basis for $\mathrm{Row}(\mathbf{A})$ and let $\mathcal{B}'$ be a basis for $\mathrm{Null}(\mathbf{A})$. Prove that

$$\mathcal{B} \cup \mathcal{B}'$$

is a basis for $\mathbb{R}^n$.

Problem 188. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(a) Write the normal equations associated with $\mathbf{A}\mathbf{x} = \mathbf{b}$.

(b) Find all least-squares solutions of $\mathbf{Ax} = \mathbf{b}$. If the least-squares solution is unique, state that it is unique.

Problem 189. (2022 Spring KAIST Introduction to Linear Algebra). Prove or disprove the following statement: if $\mathbf{P}$ is the standard matrix of an orthogonal projection of $\mathbb{R}^n$ onto a subspace of $\mathbb{R}^n$, then

$$\mathrm{tr}(\mathbf{P}) = \mathrm{rk}(\mathbf{P}).$$

Problem 190. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \end{pmatrix}.$$

Does $\mathbf{A}$ have full column rank? If not, find a basis for $\mathrm{Col}(\mathbf{A})$ consisting of columns of $\mathbf{A}$.

Problem 191. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 2 \\ 0 & 0 & -3 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix}.$$

(a) Find all eigenvalues of $\mathbf{A}$ together with their algebraic multiplicities.

(b) Determine whether $\mathbf{A}$ is orthogonally diagonalizable. If it is, find an orthonormal basis of $\mathbb{R}^4$ consisting of eigenvectors of $\mathbf{A}$; otherwise, explain why not.

(c) Find the solution set of

$$\mathbf{Ax} = \begin{pmatrix} 200 \\ 150 \\ 300 \\ -75 \end{pmatrix}.$$

Problem 192. (2022 Spring KAIST Introduction to Linear Algebra). Show that a square matrix is nilpotent if and only if 0 is its only eigenvalue.

Problem 193. (2022 Spring KAIST Introduction to Linear Algebra). Find a column-row factorization (CR decomposition) of each matrix.

(a)

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{pmatrix}.$$

(b)

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & -1 \end{pmatrix}.$$

Problem 194. (2022 Spring KAIST Introduction to Linear Algebra). Let W be a plane through the origin in $\mathbb{R}^3$.

(a) Show that the standard matrix of the orthogonal projection of $\mathbb{R}^3$ onto W is symmetric.

(b) Find the standard matrix $\mathbf{P}$ of the orthogonal projection onto the plane

$$x - y + z = 0.$$

Problem 195. (2022 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}.$$

(a) Find a QR decomposition of $\mathbf{A}$.

(b) Find the least-squares solution $\hat{\mathbf{x}}$ of

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Problem 196. (2022 Spring KAIST Introduction to Linear Algebra). Prove the following statements.

(a) Let $S \subseteq \mathbb{R}^n$ be a set of nonzero, pairwise orthogonal vectors. Show that S is linearly independent.

(b) If $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are eigenvectors of a matrix $\mathbf{A}$ corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$, show that

$$\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$$

is linearly independent.

Problem 197. (2022 Spring KAIST Introduction to Linear Algebra).

(a) Let W be a subspace of $\mathbb{R}^n$, and define

$$U = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \perp \mathbf{w} \text{ for every } \mathbf{w} \in W\}.$$

Show that U is a subspace of $\mathbb{R}^n$.

(b) Let $W \subseteq \mathbb{R}^5$ be spanned by

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 2 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 1 \\ 1 \\ -2 \\ 0 \\ -1 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Find a basis for the subspace U defined in part (a).

Problem 198. (2022 Spring KAIST Introduction to Linear Algebra). Suppose an invertible square matrix $\mathbf{A}$ has a singular value decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}.$$

Give singular value decompositions of $\mathbf{A}^{\mathsf{T}}$ and $\mathbf{A}^{-1}$ in terms of $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}$.

Problem 199. (2022 Spring KAIST Introduction to Linear Algebra). Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}.$$

Problem 200. (2022 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A}$ be a singular symmetric $n \times n$ matrix, and let $\mathbf{B}$ be a symmetric perturbation of $\mathbf{A}$. An eigenvector corresponding to the largest eigenvalue is called a dominating eigenvector. The MATLAB code below computes the angle between dominating eigenvectors of $\mathbf{A}$ and $\mathbf{B}$, and the angle between $\mathrm{Col}(\mathbf{A})$ and the dominating eigenvector of $\mathbf{B}$. Fill in blanks (1)–(9).

```
% Input:  a (singular) symmetric n by n matrix A.
n = size(A,1);
E = ___(1)___; % Generate an n by n matrix with random numbers
    % from a standard normal distribution.
epsilon = 0.001;
B = A + epsilon*1/2*(E+E');
% Compute vA and vB that are eigenvectors
% corresponding to the largest eigenvalues of A and B, respectively.
vA = final_MAS109(A, 1); vB = final_MAS109(B, 1);
% Compute cos(theta), where theta is an angle between vA and vB.
cos_theta = ___(2)___;
% Compute CA whose columns are orthonormal eigenvectors of A
% that span the column space of A.
CA = ___(3)___;
% Express vB as a sum of two orthogonal terms vB1 and vB2.
vB1 = ___(4)___; % projection of vB onto the column space of A
vB2 = ___(5)___; % projection of vB onto the null space of A
% Compute cos(phi), where phi is an angle between vB and the column space of A.
cos_phi = ___(6)___;
% Define final_MAS109, which outputs CA, whose columns are orthonormal
% eigenvectors corresponding to the k largest eigenvalues of A.
function ___(7)___
    [V, D] = eig(A); % for symmetric A, eigenvectors are orthonormal
    lambdas = ___(8)___; % eigenvalues in vector form
    [sorted_lambdas index] = sort(lambdas, 'descend');
    % Example:  if lambdas = [10 -5 4], then index = [1 3 2].
    CA = ___(9)___;
end
```

Problem 201. (2022 Fall KAIST Introduction to Linear Algebra).

(a) Let P_1 be the plane through the points

$$(1, 3, -3), \qquad (8, 1, 6), \qquad (0, 5, -6).$$

Find an equation for P_1.

(b) Let P_2 be the plane with normal vector $(11, -3, 5)$ passing through $(0, 12, 11)$. Show that $P_1 \cap P_2$ is a line, and give a vector equation for this line.

Problem 202. (2022 Fall KAIST Introduction to Linear Algebra).

(a) Find an LU decomposition of

$$\mathbf{A} = \begin{pmatrix} 2 & 8 & 2 \\ -4 & -17 & -2 \\ 6 & 23 & 11 \end{pmatrix}.$$

(b) Use the LU decomposition from part (a) to solve

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Problem 203. (2022 Fall KAIST Introduction to Linear Algebra). For which values of a does the system

$$\begin{aligned} 3x_1 - 8x_2 - 5x_3 + 10x_4 &= 9, \\ -x_1 + x_2 - 3x_3 - 6x_4 &= -2, \\ -2x_1 + 8x_2 + 20x_3 + 16x_4 &= -3, \\ x_1 - x_2 + 3x_3 + a(a-1)x_4 &= a - 1 \end{aligned}$$

have no solution, exactly one solution, or infinitely many solutions?

Problem 204. (2022 Fall KAIST Introduction to Linear Algebra). Use elementary row operations to find the inverse of

$$\mathbf{A} = \begin{pmatrix} 6 & 18 & 3 & 0 \\ 16 & 33 & 8 & -2 \\ 1 & 3 & 0 & 0 \\ -8 & -16 & -4 & 1 \end{pmatrix}.$$

Problem 205. (2022 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be nilpotent; that is, $\mathbf{A}^k = \mathbf{0}$ for some integer $k > 0$.

(a) Show that the characteristic polynomial of $\mathbf{A}$ is λ^n; equivalently,

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = \lambda^n.$$

(b) Prove or disprove that $3\mathbf{I}_n + \mathbf{A}^2$ is invertible. If it is invertible, find its inverse.

Problem 206. (2022 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{200 \times 100}$ satisfy

$$\|\mathbf{A}\mathbf{x}\| = \|\mathbf{x}\| \qquad \text{for every } \mathbf{x} \in \mathbb{R}^{100}.$$

Using direct computation, show that the columns of $\mathbf{A}$ are orthonormal.

Problem 207. (2022 Fall KAIST Introduction to Linear Algebra).

(a) Let $\mathbf{A} \in \mathbb{R}^{9 \times 9}$ satisfy $\det(\mathbf{A}) = 9$. Find

$$\det\big((\operatorname{adj}(\mathbf{A}))^{-1}\big).$$

(b) Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be invertible and have integer entries. Prove that every entry of $\mathbf{B}^{-1}$ is an integer if and only if

$$\det(\mathbf{B}) = \pm 1.$$

Problem 208. (2022 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 0 & 3 \\ 3 & 0 & 0 \end{pmatrix}.$$

(a) Find all complex eigenvalues and eigenvectors of $\mathbf{A}$.

(b) Express the operator

$$T_{\mathbf{A}} \colon \mathbb{R}^3 \to \mathbb{R}^3, \qquad T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x},$$

as a composition of special operators. Describe all relevant properties using, as applicable, the terms contraction, dilation, expansion, orthogonal, rotation, reflection, and shearing. For a rotation, state the axis and angle; for a reflection, state which vectors are reflected and which are fixed.

Problem 209. (2022 Fall KAIST Introduction to Linear Algebra). Let $T \colon \mathbb{R}^4 \to \mathbb{R}^4$ be a linear operator, and let $\mathbf{v}_1, \dots, \mathbf{v}_4, \mathbf{w}_1, \dots, \mathbf{w}_4 \in \mathbb{R}^4$. Define

$$\mathbf{u}_i = \begin{pmatrix} T(\mathbf{v}_i) \cdot \mathbf{w}_1 \\ T(\mathbf{v}_i) \cdot \mathbf{w}_2 \\ T(\mathbf{v}_i) \cdot \mathbf{w}_3 \\ T(\mathbf{v}_i) \cdot \mathbf{w}_4 \end{pmatrix}, \qquad i = 1, 2, 3, 4.$$

Show that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4$ are linearly independent if and only if T is injective and both

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \qquad \text{and} \qquad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$$

are linearly independent.

Hint. Use the matrices

$$\mathbf{V} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \end{pmatrix}, \qquad \mathbf{W} = \begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 \end{pmatrix},$$

and the matrix representation $[T]$.

Problem 210. (2022 Fall KAIST Introduction to Linear Algebra). Let W be the plane through the origin in $\mathbb{R}^3$ perpendicular to

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},$$

and let $\mathbf{A}$ be the standard matrix of reflection about W.

(a) Find linearly independent vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ such that

$$\mathbf{A}\mathbf{v} = \mathbf{v}, \qquad \mathbf{A}\mathbf{w} = \mathbf{w}.$$

(b) Find $\mathbf{A}$.

Problem 211. (2022 Fall KAIST Introduction to Linear Algebra). Consider the MATLAB code below.

```
I = 1:0.5:2.3;
b = zeros(10, 1);
for i = 2*I
    b(i) = i^2;
```

```
end
u = [2 1 5 3];
v = [1 2];
n = numel(u);
w = zeros(n, 1);
j = 0;
for i = 1:n
    if u(i) > 2
        w(j+1) = u(i);
        j = j+1;
    end
end
w = w(1:j);
```

What are the outputs of the following commands?

(a) `disp(b(1)+b(3)+b(5))`
(b) `disp(u'*v)`
(c) `disp(u.*[v v])`
(d) `disp(w)`

Problem 212. (2023 Spring KAIST Introduction to Linear Algebra). Let L be the line through $(1,1,1)$ and $(2,4,3)$.

(a) Find a parametric equation of the plane P through $Q=(-1,2,3)$ that is perpendicular to L.
(b) Find the intersection point B of P and L, and find the distance between Q and B.

Problem 213. (2023 Spring KAIST Introduction to Linear Algebra). For $a \in \mathbb{R}$, consider the linear system

$$\begin{pmatrix} 1 & 2 & 6a \\ 4 & 1 & 4 \\ 7 & 0 & -a^2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -a \\ a^2-1 \\ -a+10 \end{pmatrix}.$$

For each value of a, determine the number of solutions of the system.

Problem 214. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 10 \\ 100 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix},$$

and let $\mathbf{e}_i^{\mathsf{T}}$ be the ith standard basis row vector of $\mathbb{R}^4$. Compute each of the following whenever it is defined.

(1) $\mathrm{tr}(\mathbf{u}\mathbf{v}^{\mathsf{T}})$.
(2) $\mathrm{tr}(\mathbf{u}^{\mathsf{T}}\mathbf{v})$.
(3) $\mathbf{e}_2^{\mathsf{T}}\mathbf{u}\mathbf{v}^{\mathsf{T}}$.
(4) $\mathbf{e}_2\mathbf{u}\mathbf{v}^{\mathsf{T}}$.
(5) $\mathbf{u}\mathbf{v}^{\mathsf{T}}\mathbf{e}_2^{\mathsf{T}}$.
(6) $\mathbf{u}\mathbf{v}^{\mathsf{T}}\mathbf{e}_2$.

Problem 215. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{3\times 3}$ be invertible and satisfy

$$\mathbf{A}^2 = 2\mathbf{A}.$$

Find all possible values of $\det(\mathbf{A})$, and justify both the possible and impossible values.

Problem 216. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{e}_i$ be the ith standard basis vector of $\mathbb{R}^n$. For arbitrary linear operators $T_1, T_2 \colon \mathbb{R}^n \to \mathbb{R}^n$, determine which of the following functions are always linear. Justify your answers.

(1) $f_1(\mathbf{x}) = T_1(\mathbf{x}) + T_2(\mathbf{x})$.
(2) $f_2(\mathbf{x}) = T_1(\mathbf{x}) + \mathbf{e}_1$.
(3) $f_3(\mathbf{x}) = T_1(\mathbf{x}) \cdot T_2(\mathbf{x})$.
(4) $f_4(\mathbf{x}) = \mathbf{e}_1 \cdot T_1(\mathbf{x})$.

Problem 217. (2023 Spring KAIST Introduction to Linear Algebra). Let $T \colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator. Prove that the following statements are equivalent:

(i) $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{R}^n$.
(ii) $T(\mathbf{x}) \cdot T(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Then use this result to show that every length-preserving linear operator $T \colon \mathbb{R}^n \to \mathbb{R}^n$ preserves angles.

Problem 218. (2023 Spring KAIST Introduction to Linear Algebra). Determine whether each statement is true or false. A false statement should be disproved by a counterexample.

(1) An $n \times n$ matrix $\mathbf{A}$ is invertible if and only if a row echelon form of $\mathbf{A}$ obtained by Gaussian elimination is upper triangular with every diagonal entry equal to 1.
(2) Every subspace $W \subseteq \mathbb{R}^n$ contains the zero vector.
(3) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric $n \times n$ matrices, then $\mathbf{AB}$ is symmetric.
(4) For a fixed nonzero vector $\mathbf{v} \in \mathbb{R}^n$, the function $T \colon \mathbb{R}^n \to \mathbb{R}^n$ defined by $T(\mathbf{x}) = \mathbf{x} + \mathbf{v}$ is a linear operator.
(5) If $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $n > m$, then $\mathrm{Null}(\mathbf{A})$ contains a nonzero vector.

Problem 219. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}.$$

(a) Determine whether $\mathbf{A}$ admits an LU decomposition. If it does, find one; otherwise, find a PLU decomposition.
(b) Compute $\det(\mathbf{A})$.

Problem 220. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ with $n \geq 2$. Express

$$\det(\mathrm{adj}(\mathbf{A}))$$

in terms of n and $\det(\mathbf{A})$.

Problem 221. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 1 \\ 2 & -3 & -2 \\ -1 & 2 & 2 \end{pmatrix}.$$

(a) Find all eigenvalues of $\mathbf{A}$, their algebraic multiplicities, and the corresponding eigenvectors.
(b) Find $3\mathbf{A}^{-1}$ if it exists.

Problem 222. (2023 Spring KAIST Introduction to Linear Algebra). A square matrix $\mathbf{A}$ is called idempotent if $\mathbf{A}^2 = \mathbf{A}$.

(a) Determine all possible eigenvalues of an idempotent matrix. For each possible eigenvalue, give an example of an idempotent matrix having that eigenvalue.
(b) Can a single idempotent matrix have all of the possible eigenvalues found in part (a)? If so, give an example.

Problem 223. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \frac{1}{3}\begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}.$$

(a) Show that $\mathbf{A}$ is orthogonal.
(b) Find all eigenvalues of $\mathbf{A}$ and the eigenspace corresponding to each eigenvalue.
(c) Determine whether the orthogonal linear operator represented by $\mathbf{A}$ is a rotation about a line through the origin, a reflection about a plane through the origin, or neither. In the rotation case, find a direction vector of the axis and the rotation angle. In the reflection case, find a normal vector of the reflection plane.

Problem 224. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{3\times 3}$ be invertible.

(a) Determine all $c \in \mathbb{R}$ for which

$$\mathbf{XY} - \mathbf{YX} = c\mathbf{I}_3$$

can hold. For each possible value, give an example of $\mathbf{X}$ and $\mathbf{Y}$; for each impossible value, explain why it is impossible.
(b) Determine all $c \in \mathbb{R}$ for which

$$\mathbf{XYX}^{-1}\mathbf{Y}^{-1} = c\mathbf{I}_3$$

can hold. For each possible value, give an example of $\mathbf{X}$ and $\mathbf{Y}$; for each impossible value, explain why it is impossible.

Problem 225. (2023 Spring KAIST Introduction to Linear Algebra). The following commands are entered in the MATLAB Command Window.

```
A = magic(4);
A_times_ones = A * [1; 1; 1; 1];
sum_row2 = A_times_ones(2);
disp(sum_row2);
```

The MATLAB function `magic` takes a positive integer n and returns an $n \times n$ matrix containing the integers $1, 2, \ldots, n^2$ such that every row sum and every column sum is the same.

(a) What is the output of the code? Explain why.

(b) Assuming the code above has already been executed, write MATLAB code that creates a variable `B` whose value is the product of `A` with itself.

(c) Assuming the code from part (b) has also been executed, the following commands are run:

```
B_times_ones = B * [1, 1, 1, 1]';
disp(B_times_ones);
```

A column vector whose entries are all 1156 is displayed. Explain why. *Hint.* $1156 = 34^2$.

Now suppose the following MATLAB function is available:

```
function y = my_func(x)
    if x > 0
        y = sqrt(x);
    elseif x < 0
        y = x^2;
    else
        y = 0;
    end
end
```

(d) Assuming all code through part (c) has already been executed, what is the output of

```
x = [1, 1, 1, 1] * B_times_ones;
disp(my_func(-3) + my_func(x));
```

Problem 226. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{p}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{p}_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{p}_3 = \begin{pmatrix} 1 \\ 5 \\ x \end{pmatrix}$$

be three points in $\mathbb{R}^3$. For which values of x does there exist a two-dimensional linear subspace containing all three points? Justify your answer.

Problem 227. (2023 Spring KAIST Introduction to Linear Algebra). Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator, and let

$$\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq \mathbb{R}^n$$

be linearly independent. Show that if

$$\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)\}$$

is not linearly independent, then T is not injective. Equivalently, show that there exist distinct $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ such that

$$T(\mathbf{x}) = T(\mathbf{y}).$$

Problem 228. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A}$ be row equivalent to

$$\mathbf{R} = \begin{pmatrix} 0 & 1 & 0 & 0 & 4 & 0 \\ 0 & 0 & 1 & 1 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

For each item, either answer the question or state that there is not enough information.

(a) Find $\dim \mathrm{Null}(\mathbf{A})$.
(b) Find a basis for $\mathrm{Null}(\mathbf{A})$.
(c) Find $\dim \mathrm{Col}(\mathbf{A})$.
(d) Find a basis for $\mathrm{Col}(\mathbf{A})$.
(e) Find $\dim \mathrm{Row}(\mathbf{A})$.
(f) Find a basis for $\mathrm{Row}(\mathbf{A})$.

Problem 229. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathcal{B} = \left\{ \mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 2 \end{pmatrix}, \mathbf{b}_3 = \begin{pmatrix} 5 \\ 2 \\ 4 \\ 1 \end{pmatrix} \right\}$$

be a basis for a subspace $W \subseteq \mathbb{R}^4$.

(a) Find an orthogonal basis for W that contains $\mathbf{b}_1$ and whose vectors all have integer coordinates.
(b) Find a basis for $W^\perp$.
(c) Find the orthogonal projection onto W of

$$\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}.$$

Problem 230. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$V = \mathrm{Null}(\mathbf{A}) \subseteq \mathbb{R}^n$$

for some matrix $\mathbf{A}$, and suppose $\mathbf{v}_1, \ldots, \mathbf{v}_k \in V$ are linearly independent. Form a matrix $\mathbf{B}$ by appending the rows

$$\mathbf{v}_1^\mathsf{T}, \ldots, \mathbf{v}_k^\mathsf{T}$$

to the bottom of $\mathbf{A}$. If

$$\{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$$

is a basis for $\mathrm{Null}(\mathbf{B})$, prove that

$$\{\mathbf{v}_1, \ldots, \mathbf{v}_k, \mathbf{w}_1, \ldots, \mathbf{w}_m\}$$

is a basis for V.

Problem 231. (2023 Spring KAIST Introduction to Linear Algebra). Find an orthogonal diagonalization of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}.$$

Problem 232. (2023 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 6 \\ 7 \end{pmatrix}.$$

(a) Find all least-squares solutions of $\mathbf{Ax} = \mathbf{b}$.

(b) Find the least-squares solution $\mathbf{y}$ of $\mathbf{Ax} = \mathbf{b}$ with minimum norm $\|\mathbf{y}\|$.

Problem 233. (2023 Spring KAIST Introduction to Linear Algebra). Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}.$$

Problem 234. (2023 Spring KAIST Introduction to Linear Algebra). Consider

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} d & e & f \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) For which values of a, b, c, d, e, f are $\mathbf{A}$ and $\mathbf{B}$ orthogonal matrices representing rotations?

(b) When both matrices are orthogonal and represent rotations, let

$$\mathbf{C} = \mathbf{A}^{\mathsf{T}}\mathbf{B}\mathbf{A}.$$

Show that $\mathbf{C}$ represents a rotation about a line through the origin in $\mathbb{R}^3$, and find its axis and angle. You need not determine the orientation of the axis or the direction of rotation.

Problem 235. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ have rank 1, where $n > 1$.

(a) Show directly, without invoking the corresponding rank-factorization theorem, that there exist nonzero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ such that

$$\mathbf{A} = \mathbf{u}\mathbf{v}^{\mathsf{T}}.$$

(b) Find all eigenvalues of $\mathbf{A}$, together with their algebraic multiplicities, in terms of $\mathbf{u}$ and $\mathbf{v}$. You may need to consider two cases.

(c) Give a necessary and sufficient condition, in terms of $\mathbf{u}$ and $\mathbf{v}$, for $\mathbf{A} = \mathbf{u}\mathbf{v}^{\mathsf{T}}$ to be diagonalizable.

Problem 236. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$. Suppose $\mathcal{B}_1$ and $\mathcal{B}_2$ are orthonormal bases for $\mathbb{R}^n$, where $\mathcal{B}_1$ consists of eigenvectors of $\mathbf{A}$ and $\mathcal{B}_2$ consists of eigenvectors of $\mathbf{B}$. Show that there exists an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A} + \mathbf{B}$.

Problem 237. (2023 Spring KAIST Introduction to Linear Algebra). Let $Q(\mathbf{x})$ be a quadratic form on $\mathbb{R}^3$, where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

(a) Show that there exists a rotation matrix $\mathbf{R} = \mathbf{R}_{\mathbf{u},\theta}$, depending on Q, such that $Q(\mathbf{Rx})$ is diagonal; that is,

$$Q(\mathbf{Rx}) = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \lambda_3 x_3^2$$

for some $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$.

(b) Let

$$Q(\mathbf{x}) = -x_1^2 + x_2^2 + 4x_1x_3 + 4x_2x_3.$$

Find a rotation matrix $\mathbf{R}$ such that $Q(\mathbf{Rx})$ is diagonal. You need not determine the axis or angle of rotation.

Problem 238. (2023 Spring KAIST Introduction to Linear Algebra). Find all $n \times n$ upper triangular orthogonal matrices.

Problem 239. (2023 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be invertible. Show that there exist a unique orthogonal matrix $\mathbf{Q}$ and a unique upper triangular matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{QR}$$

and every diagonal entry of $\mathbf{R}$ is positive.

Problem 240. (2023 Spring KAIST Introduction to Linear Algebra). The following commands are entered in the MATLAB Command Window.

```
A = zeros(7);
for i = [1, 3, 5, 7]
    for j = [1, 3, 5, 7]
        A(i, j) = 1;
    end
end
for i = [2, 4, 6]
    for j = [2, 4, 6]
        A(i, j) = 1;
    end
end
```

(a) Some columns of `A` are eigenvectors of $\mathbf{A}$. Identify those columns and compute the corresponding eigenvalues.

(b) Suppose the following commands are then executed:

```
eigvals = eig(A);
disp(max(eigvals));
```

Here `max` returns the maximum element of its input. What is displayed? Explain why.

Suppose the MATLAB function `householder`, which takes a column vector `x`, is defined by

```
function Q = householder(x)
    [n, m] = size(x);
    if x' * x == 0
        Q = eye(n);
    else
        Q = eye(n) - 2 * (x * x') / (x' * x);
    end
end
```

Assuming all previous code has already been executed, the following commands are then run:

```
H = householder([-1; 0; 1; 0; -1; 0; 1]);
B = H * A * H;
syms t;
y = subs(det(t * eye(7) - B), t, 1);
[L, U, P] = lu(A);
[Q, R] = qr(U);
v = diag(inv(P) * L * inv(Q) * R);
```

(c) What is the value of `y` after executing the code?
(d) What is the value of `v` after executing the code?

Problem 241. (2024 Spring KAIST Introduction to Linear Algebra). Determine whether each of the following sets is a subspace of $\mathbb{R}^4$. No justification is required.

(1) $\mathrm{Span}((1,2,3,4),(0,2,0,1)) \cup \mathrm{Span}((1,0,3,0))$.
(2) $\mathrm{Span}((1,1,3,-2),(2,3,1,-1)) \cup \mathrm{Span}((1,3,-7,4))$.
(3) The set of vectors in $\mathbb{R}^4$ whose entries sum to 0.
(4) The set of vectors in $\mathrm{Span}((1,1,1,-1),(-1,1,1,1))$ whose entries sum to 100.
(5) The set of vectors in $\mathbb{R}^4$ that are orthogonal to $(1,2,3,-4)$ and whose entries sum to 0.

Problem 242. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{3\times 3}$ satisfy

$$(2\mathbf{A}^{\mathsf{T}})^{-1} = \begin{pmatrix} 3 & 0 & 2 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}.$$

(a) Find $\mathbf{A}$.
(b) Find all eigenvalues of $\mathbf{A}$, their algebraic multiplicities, and the corresponding eigenspaces.

Problem 243. (2024 Spring KAIST Introduction to Linear Algebra). For $a \in \mathbb{R}$, consider the linear system

$$\begin{pmatrix} 1 & -2 & 4 \\ -2 & 2 & -3a \\ a & -8 & a^2+2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a-1 \\ -7 \\ a \end{pmatrix}.$$

(a) Determine the number of solutions for each $a \in \mathbb{R}$.
(b) Use Cramer's rule to solve the system when $a = 0$. Do not use another method.

Problem 244. (2024 Spring KAIST Introduction to Linear Algebra). Define $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$T(a,b,c,d) = (y_1, y_2, y_3, y_4),$$

where

$$\begin{pmatrix} y_1 & y_2 \\ y_3 & y_4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

(a) Show that T is a linear transformation.
(b) Find $\ker T$ and $\det([T]_{\mathcal{E}})$, where $\mathcal{E}$ is the standard ordered basis of $\mathbb{R}^4$.

Problem 245. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{pmatrix}, \qquad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}.$$

(a) Find an LU decomposition of $\mathbf{A}$.
(b) Find all $\mathbf{v} \in \mathbb{R}^3$ such that $\mathbf{v} - (3,4,7)$ lies in

$$\mathrm{Span}(\mathbf{v}_1)^{\perp} \cap \mathrm{Span}(\mathbf{v}_2)^{\perp} \cap \mathrm{Span}(\mathbf{v}_3)^{\perp}.$$

Problem 246. (2024 Spring KAIST Introduction to Linear Algebra). Suppose $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \in \mathbb{R}^{3\times 3}$ satisfy

$$\mathbf{AD} = \begin{pmatrix} -1 & -5 & -8 \\ -1 & -5 & -8 \\ -2 & -10 & 75 \end{pmatrix}, \qquad \mathbf{BD} = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 20 & 33 \\ 0 & 0 & -46 \end{pmatrix},$$

and

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{I}_3.$$

Find a matrix in reduced row echelon form that is row equivalent to $\mathbf{D}$.

Problem 247. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{3\times 3}$ be orthogonal. Determine whether each statement is true or false, and justify your answer by proof or counterexample.

(1) $\mathbf{AB}$ is orthogonal, where

$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

(2) $|\lambda| = 1$ for every eigenvalue λ of $\mathbf{A}$.

Problem 248. (2024 Spring KAIST Introduction to Linear Algebra). On $\mathbb{R}^2$, let T_1 be reflection across the x-axis and let T_2 be reflection across the line $y = x$.

(a) Show, using their standard matrices, that $T_1 \circ T_2 \neq T_2 \circ T_1$.
(b) Describe $T_1 \circ T_2$ and $T_2 \circ T_1$ as rotations about the origin.

Problem 249. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{2\times 2}$ be symmetric. Determine whether each statement is true or false, and justify your answer by proof or counterexample.

(1) $\mathbf{A}$ has a real eigenvalue.
(2) If $\mathbf{A}$ has distinct eigenvalues λ_1, λ_2 and $\mathbf{x}_i$ is an eigenvector corresponding to λ_i for $i = 1, 2$, then $\mathbf{x}_1 \perp \mathbf{x}_2$.

Problem 250. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ and suppose that an invertible matrix $\mathbf{Q} \in \mathbb{R}^{n\times n}$ satisfies

$$\mathbf{A} = \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}.$$

Determine whether each statement is true or false, and justify your answer by proof or counterexample.

(1) $\operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\mathbf{B})$.
(2) If λ is an eigenvalue of $\mathbf{A}$, then λ is an eigenvalue of $\mathbf{B}$.
(3) If $\mathbf{x}$ is an eigenvector of $\mathbf{A}$, then $\mathbf{x}$ is an eigenvector of $\mathbf{B}$.

Problem 251. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} -2 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 \\ -7 & 4 & -1 & 0 \\ 1 & -8 & 13 & 1 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 6 & -2 & 7 \\ 0 & -1 & 11 & 5 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & -2 \end{pmatrix}.$$

Find $\det(\operatorname{adj}(\mathbf{AB}))$.

Problem 252. (2024 Spring KAIST Introduction to Linear Algebra).

(a) Show that there is no invertible skew-symmetric 3×3 real matrix.

(b) Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ is skew-symmetric and $\mathbf{I}_n$ is the identity matrix. Show that $\mathbf{I}_n - \mathbf{A}$ is invertible.

Problem 253. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{v}_1 = (1, 1, -1), \quad \mathbf{v}_2 = (2, 2, -2), \quad \mathbf{v}_3 = (2, 2, 2),$$
$$\mathbf{v}_4 = (2, 2, -6), \quad \mathbf{v}_5 = (3, 3, 5), \quad \mathbf{v}_6 = (3, 4, 2).$$

Find all $i \in \{2, 3, 4, 5, 6\}$ such that $\mathbf{v}_i$ can be expressed as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}$. Justify your answer.

Problem 254. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ have characteristic polynomial

$$p(x) = x^2 - x + 1.$$

Compute

$$\operatorname{tr}\left(\mathbf{A}^{109} + \left(\mathbf{A}^{-1}\right)^{109}\right).$$

Problem 255. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 3 & 6 & 1 \end{pmatrix}.$$

(a) Find a basis for $\operatorname{Col}(\mathbf{A})$.

(b) Write

$$\mathbf{A} = \mathbf{c}_1\mathbf{r}_1 + \mathbf{c}_2\mathbf{r}_2 + \cdots + \mathbf{c}_k\mathbf{r}_k,$$

where $k = \operatorname{rk}(\mathbf{A})$, each $\mathbf{c}_i$ is a column of $\mathbf{A}$, and each $\mathbf{r}_i$ is a 1×3 row vector.

Problem 256. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(a) Find a QR decomposition of $\mathbf{A}$.

(b) Find the least-squares solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Problem 257. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$W = \operatorname{Span}\big((1, -1, 1, 0), (0, 1, 1, -1)\big) \subseteq \mathbb{R}^4, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Find $\widehat{\mathbf{w}} \in W$ such that

$$\|\mathbf{b} - \widehat{\mathbf{w}}\| \leq \|\mathbf{b} - \mathbf{w}\|$$

for every $\mathbf{w} \in W$.

Problem 258. (2024 Spring KAIST Introduction to Linear Algebra). Find the least-squares line of best fit to the four points

$$(0,-1), \quad (-1,-1), \quad (1,1), \quad (2,2).$$

Problem 259. (2024 Spring KAIST Introduction to Linear Algebra). Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 \\ 0 & -2 & 1 \end{pmatrix}.$$

Problem 260. (2024 Spring KAIST Introduction to Linear Algebra). Let the ordered bases $\mathcal{B}_1$ and $\mathcal{B}_2$ of $\mathbb{R}^3$ be

$$\mathcal{B}_1 = \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right),$$

$$\mathcal{B}_2 = \left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix} \right).$$

Suppose $\mathcal{B}_3$ is an ordered basis of $\mathbb{R}^3$ satisfying

$$\mathbf{P}_{\mathcal{B}_2 \to \mathcal{B}_3} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

(a) Find $\mathbf{P}_{\mathcal{B}_3 \to \mathcal{B}_1}$.

(b) Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be defined by

$$T(a,b,c,d) = (a+b, c-a, b+d).$$

Find $[T]_{\mathcal{B}_1,\mathcal{E}}$, where $\mathcal{E}$ is the standard ordered basis of $\mathbb{R}^4$.

Problem 261. (2024 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & y \\ 0 & 1 & y \\ 2 & 2 & y-3 \end{pmatrix}, \qquad y \in \mathbb{R}.$$

(a) Show that 1 is an eigenvalue of $\mathbf{A}$ for every $y \in \mathbb{R}$.

(b) Find all $y \in \mathbb{R}$ for which $\mathbf{A}$ is not diagonalizable. Justify your answer.

(c) Find all $y \in \mathbb{R}$ for which $\mathbf{A}$ is orthogonally diagonalizable. Justify your answer and, in each case, orthogonally diagonalize $\mathbf{A}$.

Problem 262. (2024 Spring KAIST Introduction to Linear Algebra). Suppose $\mathbf{B} \in \mathbb{R}^{4\times 4}$ satisfies

$$\mathbf{AB} = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 3 & 6 & -2 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & -1 & 1 & 1 \\ 3 & 2 & -1 & 1 \\ -1 & 3 & 2 & -1 \\ 1 & -1 & 3 & 2 \end{pmatrix}.$$

Find a basis for $\mathrm{Row}(\mathbf{B})$ and a basis for $\mathrm{Null}(\mathbf{B}^{\mathsf{T}})$. You may use $\det(\mathbf{A}) = 45$.

Problem 263. (2024 Spring KAIST Introduction to Linear Algebra). For

$$\mathbf{A} = \begin{pmatrix} 3 & 4 & 5 & -2 \\ 6 & 8 & 10 & -4 \\ 1 & 0 & 2 & 2 \\ 2 & 4 & 3 & -4 \end{pmatrix},$$

find matrices $\mathbf{D}$ and $\mathbf{S}$ such that

$$\mathbf{A} = \mathbf{SD},$$

the rows of $\mathbf{D}$ form a linearly independent subset of the rows of $\mathbf{A}$, and $\mathbf{S}^\mathsf{T}$ is in reduced row echelon form.

Problem 264. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{P} \in \mathbb{R}^{n\times n}$ satisfy $\mathbf{P}^2 = \mathbf{P}$. Determine whether each statement is true or false, and justify your answer by proof or counterexample.

(a) $\mathbf{P}$ is diagonalizable.

(b) $\mathrm{tr}(\mathbf{P}) = \mathrm{rk}(\mathbf{P})$.

(c) If $\mathbf{P}$ is symmetric, then $\mathbf{P}$ is the standard matrix of the orthogonal projection of $\mathbb{R}^n$ onto $\mathrm{Col}(\mathbf{P})$.

Problem 265. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times k}$, and let $\mathbf{R}$ and $\mathbf{S}$ be their reduced row echelon forms, respectively. Determine whether each statement is true or false, and justify your answer by proof or counterexample.

(a) $\mathrm{rk}(\mathbf{AB}) = \mathrm{rk}(\mathbf{RB})$.

(b) $\mathrm{rk}(\mathbf{AB}) = \mathrm{rk}(\mathbf{AS})$.

Problem 266. (2024 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{4\times 5}$, $\mathbf{v}_1 \in \mathbb{R}^5$, and $\mathbf{v}_2 \in \mathbb{R}^4$. Suppose there exist linearly independent vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 \in \mathbb{R}^4$ such that

$$\mathbf{Av}_1 = \mathbf{x}_1, \qquad \mathbf{AA}^\mathsf{T}\mathbf{v}_2 = \mathbf{x}_2, \qquad \mathbf{A}^\mathsf{T}\mathbf{x}_3 = \mathbf{0}, \qquad \mathbf{AA}^\mathsf{T}\mathbf{x}_4 = \mathbf{0}.$$

Compute $\mathrm{rk}(\mathbf{A})$ and justify your answer.

Problem 267. (2024 Fall KAIST Introduction to Linear Algebra). Find all $a, b, c \in \mathbb{R}$ for which

$$\mathbf{A} = \begin{pmatrix} 1 & a-b+2c & 2a-b+c \\ 3 & -1 & a+c \\ 0 & -2 & 1 \end{pmatrix}$$

is symmetric.

Problem 268. (2024 Fall KAIST Introduction to Linear Algebra). Determine whether each function is a linear transformation. If it is, find its standard matrix.

(a) $T_1 \colon \mathbb{R}^3 \to \mathbb{R}^3$ is reflection about the origin $(0, 0, 0)$.

(b) $T_2 \colon \mathbb{R}^3 \to \mathbb{R}^3$ is orthogonal projection onto the plane

$$2x + 3y + 4z = 1.$$

Problem 269. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{v}_1 = (1, 4, 4), \qquad \mathbf{v}_2 = (-1, 3, 6), \qquad \mathbf{v}_3 = (-7, 0, 12)$$

in $\mathbb{R}^3$.

(a) Determine whether $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent. If they are, prove it. Otherwise, express one vector as a linear combination of the other two.

(b) Find all $a \in \mathbb{R}$ for which $\mathbf{w} = (3, 5, a) \in \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Problem 270. (2024 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A}$ be invertible.

(a) Show that $\mathbf{A}^\mathsf{T}$ is invertible and

$$(\mathbf{A}^\mathsf{T})^{-1} = (\mathbf{A}^{-1})^\mathsf{T}.$$

(b) Show that if $\mathbf{A}$ is skew-symmetric, that is, $\mathbf{A}^\mathsf{T} = -\mathbf{A}$, then $\mathbf{A}^{-1}$ is also skew-symmetric.

Problem 271. (2024 Fall KAIST Introduction to Linear Algebra). Find all $b \in \mathbb{R}$ for which the system

$$\begin{cases} x_1 - 2x_2 - x_3 = b^2, \\ -2x_1 + 3x_2 + 2x_3 = b, \\ -4x_1 + 7x_2 + 4x_3 = -1 \end{cases}$$

is consistent. For each such b, find the general solution.

Problem 272. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{pmatrix}.$$

(a) Determine whether $\mathbf{A}$ is orthogonal. Justify your answer.

(b) Find all real eigenvalues of $\mathbf{A}$ and the eigenspace corresponding to each one.

Problem 273. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 6 & 7 & -9 \\ 0 & 3 & -11 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ 107 \end{pmatrix}.$$

Find an LU decomposition of $\mathbf{A}$, and use it to solve $\mathbf{Ax} = \mathbf{b}$ by forward substitution followed by backward substitution. Do not use inverses of the factors obtained from the decomposition.

Problem 274. (2024 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be matrices such that $\mathbf{AB}$, $\mathbf{AC}$, $\mathbf{BD}$, and $\mathbf{CD}$ are defined. Determine whether each statement is true or false.

(a) If one column of $\mathbf{B}$ equals one column of $\mathbf{C}$, then one column of $\mathbf{AB}$ equals one column of $\mathbf{AC}$.

(b) If $\mathbf{B}$ and $\mathbf{C}$ are row equivalent, then $\mathbf{AB}$ and $\mathbf{AC}$ are row equivalent.

(c) If $\mathbf{B}$ and $\mathbf{C}$ are row equivalent, then $\mathbf{BD}$ and $\mathbf{CD}$ are row equivalent.

Problem 275. (2024 Fall KAIST Introduction to Linear Algebra). Let $T_1, T_2 \colon \mathbb{R}^n \to \mathbb{R}^m$ be linear transformations and let $c_1, c_2 \in \mathbb{R}$. Define

$$(c_1T_1 + c_2T_2)(\mathbf{u}) = c_1T_1(\mathbf{u}) + c_2T_2(\mathbf{u}), \qquad \mathbf{u} \in \mathbb{R}^n.$$

(a) Show that $c_1T_1 + c_2T_2$ is a linear transformation.

(b) Suppose now that $m = n$. A scalar $\lambda \in \mathbb{R}$ is called an eigenvalue of a linear operator $S \colon \mathbb{R}^n \to \mathbb{R}^n$ if there exists a nonzero vector $\mathbf{x} \in \mathbb{R}^n$ such that $S(\mathbf{x}) = \lambda\mathbf{x}$. Determine whether the following statement is true or false: if λ_i is an eigenvalue of T_i for $i = 1, 2$, then $c_1\lambda_1 + c_2\lambda_2$ is an eigenvalue of $c_1T_1 + c_2T_2$.

Problem 276. (2024 Fall KAIST Introduction to Linear Algebra). For an integer $n > 2$, define $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n\times n}$ by

$$a_{ij} = \begin{cases} r, & i = j, \\ 1, & i + 1 = j \text{ and } i \neq 2, \\ 0, & \text{otherwise}, \end{cases}$$

where $r \in \mathbb{R}$ is fixed.

(a) Find the characteristic polynomial of $\mathbf{A}$.

(b) Find all real eigenvalues of $\mathbf{A}$.

(c) Find the eigenspace $E_\lambda(\mathbf{A})$ corresponding to each real eigenvalue λ.

Problem 277. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{u} = (\sqrt{2}, \sqrt{2}, 0), \qquad \mathbf{v} = (0, 2, 0).$$

For $\theta \geq 0$, let $T_\theta \colon \mathbb{R}^3 \to \mathbb{R}^3$ be rotation about the positive x-axis through angle θ.

(a) Find $\theta \in [0, \pi]$ for which the area of the parallelogram spanned by $\mathbf{u}$ and $T_\theta(\mathbf{v})$ is maximized.

(b) For the angle found in part (a), find the standard matrix of T_θ.

(c) For the angle found in part (a), find a vector perpendicular to the parallelogram spanned by $\mathbf{u}$ and $T_\theta(\mathbf{v})$.

Problem 278. (2024 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n\times n}$ and let $i, j \in \{1, \ldots, n\}$.

(a) State the definition of the (i, j)-cofactor of $\mathbf{A}$ and of the adjugate matrix $\operatorname{adj}(\mathbf{A})$.

(b) State the cofactor expansions of $\det(\mathbf{A})$ along the ith row and the jth column.

(c) Compute $\mathbf{A}\operatorname{adj}(\mathbf{A})$ and justify your answer using suitable cofactor expansions.

Problem 279. (2024 Fall KAIST Introduction to Linear Algebra). Let $n > 1$, let $\mathbf{A} \in \mathbb{R}^{n\times n}$, and let $\mathbf{I}_n$ be the identity matrix.

(a) Show that if $\mathbf{A}$ is nilpotent, then $\mathbf{I}_n - \mathbf{A}$ is invertible.

(b) For the matrix $\mathbf{A} = (a_{ij})$ defined by

$$a_{ij} = \begin{cases} 1, & i + 1 = j, \\ 0, & \text{otherwise}, \end{cases}$$

find the index of nilpotency of $\mathbf{A}$, that is, the smallest positive integer k such that $\mathbf{A}^k = \mathbf{0}$, and find $(\mathbf{I}_n - \mathbf{A})^{-1}$. No proof is required for this part.

(c) For the matrix in part (b), find all fixed points of $\mathbf{A}$, that is, all $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{A}\mathbf{x} = \mathbf{x}$.

Problem 280. (2024 Fall KAIST Introduction to Linear Algebra). Suppose $\mathbf{u}_1, \dots, \mathbf{u}_4$ span $\mathbb{R}^4$, and suppose there exists a linear transformation $T\colon \mathbb{R}^4 \to \mathbb{R}^6$ such that

$$\begin{aligned} T(\mathbf{u}_1) &= (a_1, b_1, 0, 0, 0, 0), \\ T(\mathbf{u}_2) &= (0, a_2, b_2, 0, 0, 0), \\ T(\mathbf{u}_3) &= (0, 0, a_3, b_3, 0, 0), \\ T(\mathbf{u}_4) &= (b_4, 0, 0, a_4, 0, 0), \end{aligned}$$

where a_1, a_2, a_3, a_4 are even integers and b_1, b_2, b_3, b_4 are odd integers.

(a) Find $\ker T$.

(b) Find $\operatorname{im} T$.

(c) Find $c_1, c_2, c_3, c_4 \in \mathbb{R}$ such that

$$T\left(\sum_{i=1}^{4} c_i \mathbf{u}_i\right) = (1, 0, 0, 0, 0, 0).$$

Hint. Each c_i can be expressed as a rational function of $a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4$.

Problem 281. (2024 Fall KAIST Introduction to Linear Algebra).

(a) For a positive odd integer m, compute $\det(\mathbf{A})$ for an arbitrary $m \times m$ skew-symmetric matrix $\mathbf{A}$. Your answer should not depend on the specific matrix.

(b) For a positive integer n, let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be orthogonal with $\det(\mathbf{B}) < 0$. Compute

$$\det(\mathbf{B} + \mathbf{I}_n).$$

Your answer should not depend on the specific matrix $\mathbf{B}$.

Problem 282. (2024 Fall KAIST Introduction to Linear Algebra).

(a) Find the standard matrix $\mathbf{P}$ for the orthogonal projection of $\mathbb{R}^3$ onto the plane

$$x - 5y + 10z = 0.$$

(b) Use $\mathbf{P}$ to find the orthogonal projection of $(5, 2, 1) \in \mathbb{R}^3$ onto the plane.

Problem 283. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$W = \operatorname{Span}\big((1, 1, 0), (1, 2, -1), (0, 1, 1)\big) \subseteq \mathbb{R}^3.$$

(a) Find an orthonormal basis for W.

(b) Find a QR decomposition of

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Problem 284. (2024 Fall KAIST Introduction to Linear Algebra). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ be row equivalent, and let $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{b}_1, \dots, \mathbf{b}_n$ be their column vectors, respectively. Prove that for any $c_1, \dots, c_n \in \mathbb{R}$,

$$\sum_{i=1}^{n} c_i \mathbf{a}_i = \mathbf{0} \quad \Longleftrightarrow \quad \sum_{i=1}^{n} c_i \mathbf{b}_i = \mathbf{0}.$$

Problem 285. (2024 Fall KAIST Introduction to Linear Algebra). Let $\mathcal{B}$ be any ordered basis of $\mathbb{R}^n$. Prove that vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are linearly independent if and only if their coordinate vectors

$$[\mathbf{v}_1]_{\mathcal{B}}, \dots, [\mathbf{v}_k]_{\mathcal{B}}$$

are linearly independent in $\mathbb{R}^n$.

Problem 286. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{P} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix}.$$

Let $\mathcal{E}$ denote the standard ordered basis of $\mathbb{R}^3$.

(a) Find the ordered basis $\mathcal{B}$ of $\mathbb{R}^3$ such that $\mathbf{P}$ is the transition matrix from $\mathcal{B}$ to $\mathcal{E}$.

(b) Find the ordered basis $\mathcal{C}$ of $\mathbb{R}^3$ such that $\mathbf{P}$ is the transition matrix from $\mathcal{E}$ to $\mathcal{C}$.

Problem 287. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$W = \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5) \subseteq \mathbb{R}^4,$$

where

$$\mathbf{v}_1 = (2, 7, 3, 1), \quad \mathbf{v}_2 = (1, 0, 4, 3), \quad \mathbf{v}_3 = (0, 2, 3, 0),$$
$$\mathbf{v}_4 = (3, 6, 1, 2), \quad \mathbf{v}_5 = (1, 0, 0, 9).$$

(a) Find a subset of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ that forms a basis for W.

(b) Express each vector not chosen for the basis as a linear combination of the basis vectors.

Problem 288. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} -1 & -3 \\ 1 & 3 \\ 2 & 6 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

(a) Find all least-squares solutions of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

(b) Find the least-squares solution with minimum norm $\|\mathbf{x}\|$.

Problem 289. (2024 Fall KAIST Introduction to Linear Algebra).

(a) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ have singular value decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}.$$

For $i = 1, \dots, n$, let $\mathbf{u}_i$ and $\mathbf{v}_i$ be the ith columns of $\mathbf{U}$ and $\mathbf{V}$, respectively, and let σ_i be the (i, i)-entry of $\mathbf{\Sigma}$. Find an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$, and an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}\mathbf{A}^{\mathsf{T}}$.

(b) Find a singular value decomposition of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix}.$$

Problem 290. (2024 Fall KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 3 & -5 & -1 & 5 & 4 \\ 0 & -3 & 2 & 7 & 6 \\ 4 & -5 & -9 & 2 & -5 \\ 2 & 6 & 8 & 3 & -6 \end{pmatrix}.$$

(a) Show that

$$W = \left\{ \mathbf{b} \in \mathbb{R}^5 \mid \mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{b} \text{ has at least one solution } \mathbf{x} \in \mathbb{R}^5 \right\}$$

is a subspace of $\mathbb{R}^5$. *Hint.* Show that W is one of the fundamental spaces of a matrix.

(b) Find $\dim W$ and $\dim W^\perp$.

Problem 291. (2024 Fall KAIST Introduction to Linear Algebra). Let $W \subseteq \mathbb{R}^n$ be an m-dimensional subspace with $0 < m < n$, and let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be the orthogonal projection onto W. Show that there exists an ordered basis $\mathcal{B}$ of $\mathbb{R}^n$ such that

$$[T]_\mathcal{B} = \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

for some $m \times (n-m)$ matrix $\mathbf{A}$.

Problem 292. (2024 Fall KAIST Introduction to Linear Algebra). Consider the proposition: if square matrices $\mathbf{A}$ and $\mathbf{B}$ are similar, then their characteristic polynomials are equal.

(a) Prove the proposition.

(b) Prove or disprove its converse.

(c) Prove or disprove the following statement: if $\mathbf{A}$ and $\mathbf{B}$ are diagonalizable square matrices and have the same characteristic polynomial, then they are similar.

Hint. You may use without proof that diagonal matrices with the same characteristic polynomial are similar.

Problem 293. (2024 Fall KAIST Introduction to Linear Algebra). Find vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^3$ and $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3 \in \mathbb{R}^4$ satisfying all of the following conditions.

(i) The entries of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are exactly the twelve integers $1, 2, \ldots, 12$, each used once.

(ii) The entries of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are exactly the twelve integers $1, 2, \ldots, 12$, each used once.

(iii) For every $i \in \{1,2,3,4\}$ and $j \in \{1,2,3\}$, the vectors $\mathbf{v}_i$ and $\mathbf{w}_j$ share exactly one component.

(iv) The subspaces $\operatorname{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) \subseteq \mathbb{R}^3$ and $\operatorname{Span}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3) \subseteq \mathbb{R}^4$ have the same dimension.

Problem 294. (2024 Fall KAIST Introduction to Linear Algebra). Let n be a positive integer and let $\mathbf{I}_n$ be the identity matrix.

(a) Suppose $\mathbf{A} \in \mathbb{R}^{n\times n}$ satisfies $\mathbf{A}^2 = \mathbf{I}_n$. Prove that:

(i) the eigenvalues of $\mathbf{A}$ are contained in $\{1, -1\}$;

(ii) $\operatorname{Col}(\mathbf{A} - \mathbf{I}_n) \subseteq \operatorname{Null}(\mathbf{A} + \mathbf{I}_n)$;

(iii) $\mathbf{A}$ is diagonalizable.

(b) Suppose $\mathbf{B} \in \mathbb{R}^{n\times n}$ is orthogonal. Show that if $\mathbf{B}$ is diagonalizable over $\mathbb{R}$, then

$$\mathbf{B}^2 = \mathbf{I}_n.$$

Problem 295. (2025 Spring KAIST Introduction to Linear Algebra). Consider a system of m linear equations in n unknowns.

(a) If the system is homogeneous, explain why it has at least one solution.

(b) If $m < n$, systems often have solutions. Give an example with $m = 2$ and $n = 3$ of the form

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1, \\ a_{21}x + a_{22}y + a_{23}z = b_2, \end{cases}$$

in which the six coefficients $a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}$ are pairwise distinct and the system has no solution. You need not solve the system.

(c) If $m > n$, systems often have no solutions. Give an example with $m = 3$ and $n = 2$ of the form

$$\begin{cases} a_{11}x + a_{12}y = b_1, \\ a_{21}x + a_{22}y = b_2, \\ a_{31}x + a_{32}y = b_3, \end{cases}$$

in which the six coefficients $a_{11}, a_{12}, a_{21}, a_{22}, a_{31}, a_{32}$ are pairwise distinct and the system has a unique solution. You need not solve the system.

Problem 296. (2025 Spring KAIST Introduction to Linear Algebra). Consider the system

$$\begin{cases} x + 2y + 3z = 14, \\ 2x - y + z = 3, \\ 3x + 4y - z = 8. \end{cases}$$

(a) Write the augmented matrix and use elementary row operations to reduce it to reduced row echelon form.

(b) Determine whether the system has a unique solution, infinitely many solutions, or no solution. Justify your answer and, if solutions exist, find all of them.

Problem 297. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

(a) Compute $\det(\mathbf{A})$ and list all possible cases for the number of solutions of $\mathbf{Ax} = \mathbf{b}$ as $\mathbf{b}$ ranges over $\mathbb{R}^3$.

(b) Determine the number of free variables in $\mathbf{Ax} = \mathbf{0}$ and find the general solution in parametric form.

(c) For

$$\mathbf{x}_1 = \begin{pmatrix} -3 \\ -3 \\ -3 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix},$$

determine whether each vector belongs to the range of $T_{\mathbf{A}} \colon \mathbb{R}^3 \to \mathbb{R}^3$, $T_{\mathbf{A}}(\mathbf{x}) = \mathbf{Ax}$. Justify your answers.

Problem 298. (2025 Spring KAIST Introduction to Linear Algebra).

(a) State the Leibniz formula for the determinant of an $n \times n$ matrix $\mathbf{A} = (a_{ij})$ in terms of permutations.

(b) Using the definition from part (a), compute

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 3 & -1 \\ -1 & 2 & 0 \end{pmatrix}.$$

(c) For $1 \leq i \leq n$, state the cofactor expansion of $\det(\mathbf{A})$ along the ith row.

(d) Using cofactor expansion along suitable rows or columns, compute

$$\det \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & -1 & 4 & 5 \\ 0 & 0 & 0 & 2 & -3 \\ 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & -1 \end{pmatrix}.$$

At each step, indicate the row or column used.

(e) Using elementary row operations, find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}.$$

Problem 299. (2025 Spring KAIST Introduction to Linear Algebra). Recall that the adjugate matrix of a square matrix is the transpose of its cofactor matrix.

(a) Find $\operatorname{adj}(\mathbf{A})$ and $\mathbf{A}^{-1}$ for

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -2 & 0 & -4 \end{pmatrix}.$$

(b) Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ with $n > 1$. Prove that

$$\det(\operatorname{adj}(\mathbf{B})) = \det(\mathbf{B})^{n-1}.$$

Problem 300. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Find the reduced row echelon form of the $n \times n$ matrix

$$\begin{pmatrix} 0 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 0 & 1 & \cdots & 1 & 1 \\ 1 & 1 & 0 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & 0 & 1 \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{pmatrix}.$$

(b) Find the minimum possible number of zero entries in an invertible $n \times n$ matrix whose entries belong to $\{0, 1\}$. Justify your answer.

Problem 301. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Find the characteristic polynomial of

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 0 & -a_0 \\ 1 & 0 & 0 & 0 & -a_1 \\ 0 & 1 & 0 & 0 & -a_2 \\ 0 & 0 & 1 & 0 & -a_3 \\ 0 & 0 & 0 & 1 & -a_4 \end{pmatrix}.$$

(b) Find a matrix whose characteristic polynomial is

$$2 - 3\lambda + 4\lambda^2 - \lambda^3 + 2\lambda^4 + \lambda^5.$$

Problem 302. (2025 Spring KAIST Introduction to Linear Algebra). Let λ be an eigenvalue of $\mathbf{A} \in \mathbb{R}^{n\times n}$ with corresponding eigenvector $\mathbf{x}$.

(a) If $\mathbf{P} \in \mathbb{R}^{n\times n}$ is invertible, prove that λ is an eigenvalue of $\mathbf{P}^{-1}\mathbf{AP}$ and find a corresponding eigenvector.

(b) If $\mathbf{A}$ is invertible, prove that λ^{-1} is an eigenvalue of $\mathbf{A}^{-1}$ and find a corresponding eigenvector.

Problem 303. (2025 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{2\times 2}$ be symmetric.

(a) Prove that $\mathbf{A}$ has a real eigenvalue.

(b) If $\mathbf{A}$ has one repeated eigenvalue, prove that $\mathbf{A}$ is a scalar multiple of $\mathbf{I}_2$.

(c) If $\mathbf{A}$ has two distinct eigenvalues, prove that eigenvectors corresponding to distinct eigenvalues are orthogonal.

Problem 304. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

(a) Prove that $\mathbf{A}$ is the standard matrix of a rotation of $\mathbb{R}^2$, and find its rotation angle $\theta \in [0, \pi]$.

(b) Prove that $\mathbf{B}$ is the standard matrix of a reflection of $\mathbb{R}^2$, and find its reflection line.

(c) Determine whether $\mathbf{AB}$ is the standard matrix of a rotation or a reflection. If it is a rotation, find its angle $\theta \in [0, \pi]$; if it is a reflection, find its reflection line.

Problem 305. (2025 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$.

(a) Prove that $\mathbf{A}^\mathsf{T}\mathbf{A}$ is symmetric.

(b) If $\mathrm{tr}(\mathbf{A}^\mathsf{T}\mathbf{A}) = 0$, prove that $\mathbf{A} = \mathbf{0}$.

(c) Prove that

$$\mathrm{Null}(\mathbf{A}) = \mathrm{Null}(\mathbf{A}^\mathsf{T}\mathbf{A}).$$

Problem 306. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Prove or disprove that $\mathbf{A}$ is the standard matrix of a rotation of $\mathbb{R}^3$. If it is, find the rotation axis and rotation angle.

Problem 307. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad V = \operatorname{Span}(\mathbf{v}_1, \mathbf{v}_2) \subseteq \mathbb{R}^3.$$

(a) Determine whether $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent, and find $\dim V$.
(b) Find an orthonormal basis for V.
(c) Extend the orthonormal basis found in part (b) to an orthonormal basis for $\mathbb{R}^3$.

Problem 308. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 2 \\ 2 & 5 & 0 & 2 \end{pmatrix}.$$

(a) Compute $\operatorname{rk}(\mathbf{A})$ and $\dim \operatorname{Null}(\mathbf{A})$.
(b) Find a basis for $\operatorname{Col}(\mathbf{A})$ and a basis for $\operatorname{Null}(\mathbf{A})$.
(c) Compute $\operatorname{rk}(\mathbf{A}^\mathsf{T})$ and $\dim \operatorname{Null}(\mathbf{A}^\mathsf{T})$.
(d) Find a basis for $\operatorname{Col}(\mathbf{A}^\mathsf{T})$ and a basis for $\operatorname{Null}(\mathbf{A}^\mathsf{T})$.

Problem 309. (2025 Spring KAIST Introduction to Linear Algebra). Let $U, W \subseteq \mathbb{R}^n$ be subspaces, and define

$$U + W = \{\mathbf{u} + \mathbf{w} \mid \mathbf{u} \in U,\ \mathbf{w} \in W\}.$$

(a) Prove that $U + W$ and $U \cap W$ are subspaces of $\mathbb{R}^n$.
(b) Prove that

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

(c) For

$$U = \operatorname{Span}\left(\begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right),$$

$$W = \operatorname{Span}\left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 8 \\ 3 \\ 6 \end{pmatrix} \right) \subseteq \mathbb{R}^4,$$

compute $\dim(U + W)$ and find a basis for $U + W$.

Problem 310. (2025 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$.

(a) Prove that

$$\operatorname{Null}(\mathbf{B}) \subseteq \operatorname{Null}(\mathbf{AB}).$$

(b) Suppose $(\mathbf{v}_1, \dots, \mathbf{v}_p)$ is a basis for $\operatorname{Null}(\mathbf{B})$ and $(\mathbf{v}_1, \dots, \mathbf{v}_{p+q})$ is a basis for $\operatorname{Null}(\mathbf{AB})$. Prove that

$$\mathbf{B}\mathbf{v}_{p+1}, \dots, \mathbf{B}\mathbf{v}_{p+q}$$

are linearly independent.

(c) Prove that

$$\dim \mathrm{Null}(\mathbf{AB}) \leq \dim \mathrm{Null}(\mathbf{A}) + \dim \mathrm{Null}(\mathbf{B})$$

and

$$\mathrm{rk}(\mathbf{AB}) \geq \mathrm{rk}(\mathbf{A}) + \mathrm{rk}(\mathbf{B}) - n.$$

(d) Prove that $\mathrm{rk}(\mathbf{A}) = k$ if and only if

$$\mathbf{A} = \mathbf{u}_1\mathbf{v}_1^\mathsf{T} + \cdots + \mathbf{u}_k\mathbf{v}_k^\mathsf{T}$$

for some linearly independent vectors $\mathbf{u}_1, \ldots, \mathbf{u}_k \in \mathbb{R}^n$ and some linearly independent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^n$.

(e) Prove that

$$\mathrm{rk}(\mathbf{A} + \mathbf{B}) \leq \mathrm{rk}(\mathbf{A}) + \mathrm{rk}(\mathbf{B})$$

and

$$\dim \mathrm{Null}(\mathbf{A} + \mathbf{B}) \geq \dim \mathrm{Null}(\mathbf{A}) + \dim \mathrm{Null}(\mathbf{B}) - n.$$

Problem 311. (2025 Spring KAIST Introduction to Linear Algebra). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

(a) A least-squares solution $\hat{\mathbf{x}}$ of $\mathbf{Ax} = \mathbf{b}$ is an exact solution of $\mathbf{A}\hat{\mathbf{x}} = \widehat{\mathbf{b}}$ for some $\widehat{\mathbf{b}} \in \mathbb{R}^m$. Express $\widehat{\mathbf{b}}$ using the orthogonal projection onto one of the fundamental spaces of $\mathbf{A}$.

(b) If $\mathbf{A}$ has full column rank, determine the number of least-squares solutions of $\mathbf{Ax} = \mathbf{b}$ and justify your answer.

(c) If $\mathbf{A}$ does not have full column rank, determine the number of least-squares solutions of $\mathbf{Ax} = \mathbf{b}$ and justify your answer.

(d) Write the normal equations for $\mathbf{Ax} = \mathbf{b}$ and explain why every solution of the normal equations is a least-squares solution of $\mathbf{Ax} = \mathbf{b}$.

(e) Find all least-squares solutions of $\mathbf{Ax} = \mathbf{b}$ for

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \\ -2 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}.$$

Problem 312. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Let $\mathbf{P} \in \mathbb{R}^{n\times n}$ satisfy $\mathrm{rk}(\mathbf{P}) = k$, $\mathbf{P}^2 = \mathbf{P}$, and $\mathbf{P}^\mathsf{T} = \mathbf{P}$. Prove that $\mathbf{P}$ is the standard matrix of the orthogonal projection onto some k-dimensional subspace $W \subseteq \mathbb{R}^n$.

(b) Classify $\mathbf{P}$ from part (a) as positive semidefinite, negative semidefinite, or indefinite.

(c) Let $\mathbf{H} \in \mathbb{R}^{n\times n}$ satisfy $\mathbf{H}^\mathsf{T} = \mathbf{H}$, and suppose $\mathbf{I}_n - \mathbf{H}$ has rank 1 and has eigenvalue 2. Prove that $\mathbf{H}$ is the standard matrix of reflection across the hyperplane $\mathbf{x}^\perp$ for some nonzero $\mathbf{x} \in \mathbb{R}^n$.

(d) Classify $\mathbf{H}$ from part (c) as positive semidefinite, negative semidefinite, or indefinite.

Problem 313. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Using the Gram–Schmidt process, find a QR decomposition of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ -1 & -2 & 3 \\ 0 & 4 & 5 \end{pmatrix}.$$

(b) Using reflections (Householder matrices), find a QR decomposition of

$$\mathbf{B} = \begin{pmatrix} 0 & 3 & 1 \\ 0 & 0 & 5 \\ 1 & 1 & 1 \end{pmatrix}.$$

Problem 314. (2025 Spring KAIST Introduction to Linear Algebra). Let

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \qquad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

and let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

(a) Prove that $\mathcal{B}$ is an ordered basis for $\mathbb{R}^3$.

(b) Find the change-of-basis matrix $\mathbf{P}_{\mathcal{B}\to\mathcal{S}}$ from $\mathcal{B}$ to the standard ordered basis $\mathcal{S}$ of $\mathbb{R}^3$.

(c) For

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix},$$

compute the coordinate vector $[\mathbf{v}]_{\mathcal{B}}$.

(d) Express $\mathbf{v}$ in terms of $\mathbf{P}_{\mathcal{B}\to\mathcal{S}}$ and $[\mathbf{v}]_{\mathcal{B}}$.

Problem 315. (2025 Spring KAIST Introduction to Linear Algebra). For $\mathbf{x} \in \mathbb{R}^3$, let $\mathbf{x}'$ be obtained by rotating $\mathbf{x}$ by $45°$ counterclockwise about the z-axis, viewed from the positive z-axis toward the origin. Let $\mathbf{x}''$ be obtained by rotating $\mathbf{x}'$ by $30°$ counterclockwise about the y-axis, viewed from the positive y-axis toward the origin.

(a) Find the matrix $\mathbf{A}$ such that $\mathbf{A}\mathbf{x} = \mathbf{x}''$ for every $\mathbf{x} \in \mathbb{R}^3$.

(b) Determine whether $\mathbf{A}$ is similar to a Householder matrix.

Problem 316. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Find a singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

(b) Find a polar decomposition of

$$\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}.$$

Problem 317. (2025 Spring KAIST Introduction to Linear Algebra).

(a) Let

$$\mathbf{A} = \begin{pmatrix} -1 & 0 & 1 \\ -3 & 2 & 4 \\ 1 & -1 & -2 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 1 & -2 \\ 2 & 2 & -3 \end{pmatrix}.$$

Determine whether $\mathbf{A}$ is similar to $\mathbf{B}$. If so, find an invertible matrix $\mathbf{P}$ such that

$$\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}\mathbf{P}.$$

(b) Let

$$\mathbf{C} = \begin{pmatrix} 1 & 3 & -3 \\ 0 & 1 & 0 \\ 0 & -3 & 4 \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Determine whether $\mathbf{C}$ is similar to $\mathbf{D}$. If so, find an invertible matrix $\mathbf{Q}$ such that

$$\mathbf{C} = \mathbf{Q}^{-1}\mathbf{D}\mathbf{Q}.$$